# Weight-count compression by activation-weighted anchoring: an exploratory empirical study

## PREPRINT REVIEW DRAFT -- no venue assumed

No established global-preservation or practical-compression claim. Original FP16 readiness remains failed. Native-tokenizer perplexities are compared within models only. Partial and cancelled attempts remain evidence, not completed cohorts.

Equations use readable ASCII notation (beta, rho, lambda; dagger denotes the pseudoinverse). The accompanying standalone manuscript.tex provides exact typeset mathematical source. This PDF is independently rendered from the same Markdown and measured figures.

Version: 2026-10-14T15:09:21.922132+00:00

# Abstract

This report examines whether matched activation-weighted low-rank approximations preserve global transformer behavior. It retains weighting/anchoring controls, full-stack and composition measurements, separate BF16/FP8 numerical controls, and the user-amended hybrid-architecture panel. Tables below are generated from retained artifacts; failed/incomplete attempts remain visible. Targeted weight-count reduction is separated from likelihood, representation drift and measured execution costs. The evidence does not establish reliable global preservation or a consistent anchoring advantage over matched whitening. Promotion remains disabled; a terminal dispatcher does not imply every scientific cohort completed.

# Study status and scope

Dispatcher: completed with incomplete cohorts. Report mode: terminal orchestration report.

The amended panel uses Gemma, Qwen3.5-0.8B and LFM2.5-1.2B-Instruct; GPT-2 is excluded. Only full-attention output projections are targeted in the hybrids. Native-tokenizer PPL is compared within each model. Validation frontier ranks and primary test ranks remain separate; no selection is made using confirmation outcomes.

## Later BF16-only practical amendment

After stopping the historical FP16/INT8 practical stage, the user authorized only BF16 model testing with at most two seeds. The new practical cohort has five declared conditions: a BF16 dense teacher, and BF16 whitening/anchor models for seeds 11 and 23. Native BF16 measured-output factor preparation retains the frozen objective and rank. Historical other-format and three-seed evidence is preserved, not relabeled or pooled with this cohort. Higher-precision fitting/metric accumulation is numerical computation, not another model-format test. Two seeds cannot satisfy the original three-seed promotion gate. The stopped stage remains charged against the original shared practical cap.

## Declared numerical-control omissions

The continuation formally omits unfinished dense_fp8_e4m3 and dense_fp8_e5m2 controls. All seven saved controls, including failed gates, and the cancelled attempt remain preserved. This scope closure does not claim that the original nine-control cohort completed or that native FP8 model inference was tested.

# Mathematical methods and frozen constants

Let W in R^{o x i} and X in R^{i x n}. Write A=UV with rank at most r. The anchored objective is ||(W-A)X||_F^2+beta n||W-A||_F^2.

The replacement is the attention-output projection after head outputs are combined; attention heads themselves are not pruned. Fitting is offline, and inference applies V followed by U in place of W. Anchoring adds no runtime branch or extra factor parameters: it changes only the fitting geometry.

For each Gemma target, the dense matrix has 1152 x 1024 = 1,179,648 weights. Rank 384 stores 1152 x 384 + 384 x 1024 = 835,584 weights: 344,064 fewer, or 29.17% of that projection. Across 26 targets the reduction is 8,945,664 weights. These are targeted weight counts, not a 29.17% whole-model reduction; embeddings, MLPs and other untargeted weights remain.

Set H=XX^T+beta nI and C=WH for exact teacher outputs. For measured outputs subtract the teacher bias first: Y_b=Y-b1^T, then C=Y_bX^T+beta nW. The extra-ridge solver adds lambda I to the geometry, with lambda=rho mean(eig(H)). For G=H+lambda I, form A_full=CG^dagger and truncate A_fullG^{1/2} to rank r; multiply back by G^{dagger/2}. Thus ridge-free full-rank anchored recovery gives W on retained spectral directions, while positive extra ridge generally shrinks it.

The implementation computes weighted covariance, eigendecomposition and weighted truncation in FP64, then returns FP32 factors for the declared storage cast. It clamps negative numerical eigenvalues to zero and retains regularized eigenvalues strictly above rcond lambda_max(G). Plain SVD uses FP32. Fixed-rank diagonal ASVD uses s_j=max(mean_t|X_{jt}|,10^{-8})^{0.5}, truncates Wdiag(s), and scales back. Factors use the recorded unbalanced convention unless a run contract explicitly declares a separate convention.

The predeclared validation anchor sweep is beta={0,0.03,0.1,0.3,1}; the extra-ridge fractions are rho={0,0.001,0.01} at beta=0.1. Validation NLL ties within 1e-6 prefer less regularization. Native evaluation uses context/stride 512/256, full shifted targets including the tail, and 16 disjoint 512-token calibration chunks for seeds 11/23/37. The primary Gemma attention-output rank is 384. The amended hybrid rank is floor((17/24)*out*in/(out+in)); frontier ranks are floor(min(out,in)/divisor) for validation-only divisors2/3/4. Exact observed run constants and factor precisions follow below; missing fields remain unrecorded.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 2; match by Row ID.

| Row ID | Run | Precision | Main rank | Pseudoinverse rcond |
|---|---|---|---|---|
| R001 | architecture-replication-20261014-082807-5c73f669 | bfloat16 | shape-derived by 17/24 factor/dense fraction | 1e-07 |
| R002 | architecture-replication-20261014-093845-97378b5e | bfloat16 | shape-derived by 17/24 factor/dense fraction | 1e-07 |
| R003 | architecture-replication-20261014-103759-8d32ab84 | bfloat16 | shape-derived by 17/24 factor/dense fraction | 1e-07 |
| R004 | baselines-20261013-175255-6a00d4e8 | float16 | 384 | 1e-07 |
| R005 | composition-20261013-185512-2bffd477 | float16 | 384 | 1e-07 |
| R006 | composition-20261014-032120-486d799f | float16 | 384 | 1e-07 |
| R007 | composition-20261014-054423-dbd821ff | float16 | 384 | 1e-07 |
| R008 | practical-20261014-113642-0fe078b5 | float16 | 384 | 1e-07 |
| R009 | practical-bf16-20261014-135107-82168ad5 | bfloat16 | 384 | 1e-07 |
| R010 | precision-readiness-20261014-060840-6db37daf | float16 | 384 | 1e-07 |
| R011 | regularization-20261013-171941-804f4c2e | float16 | 384 | 1e-07 |

Panel 2 of 2; match by Row ID.

| Row ID | Run | Config beta | Config ridge fraction | Selected solver / baseline |
|---|---|---|---|---|
| R001 | architecture-replication-20261014-082807-5c73f669 | unmeasured | unmeasured | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}} |
| R002 | architecture-replication-20261014-093845-97378b5e | unmeasured | unmeasured | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}} |
| R003 | architecture-replication-20261014-103759-8d32ab84 | unmeasured | unmeasured | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}} |
| R004 | baselines-20261013-175255-6a00d4e8 | 0.1 | 0.01 | {"baseline": "SVD and fixed-rank ASVD / whitening-only components, same FP16/rank", "solver": null} |

| Row ID | Run | Config beta | Config ridge fraction | Selected solver / baseline |
|---|---|---|---|---|
| R005 | composition-20261013-185512-2bffd477 | 0.1 | 0.01 | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null} |
| R006 | composition-20261014-032120-486d799f | 0.1 | 0.01 | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null} |
| R007 | composition-20261014-054423-dbd821ff | 0.1 | 0.01 | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null} |
| R008 | practical-20261014-113642-0fe078b5 | 0.1 | 0.01 | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null} |
| R009 | practical-bf16-20261014-135107-82168ad5 | 0.03 | 0 | {"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}} |
| R010 | precision-readiness-20261014-060840-6db37daf | 0.1 | 0.01 | {"baseline": "untouched dense BF16 teacher and copied BF16 control", "solver": null} |
| R011 | regularization-20261013-171941-804f4c2e | 0.1 | 0.01 | {"baseline": "SVD and fixed-rank ASVD / whitening-only components, same FP16/rank", "solver": null} |

# Related work and scope of the planned evidence

Data-aware low-rank compression predates this study. DRONE (https://proceedings.neurips.cc/paper/2021/hash/f56de5ef149cf0aedcc8f4797031e229-Abstract.html) derives a closed-form approximation informed by layer representations in BERT [@chen2021drone]. Fisher-weighted factorization (https://arxiv.org/abs/2207.00112) uses parameter importance to address disagreement between matrix reconstruction and task accuracy [@hsu2022weighted]. These are context for the present question: low local weight error alone cannot establish preserved global computation.

ASVD (https://arxiv.org/abs/2312.05821) adapts decomposition to activation distributions and layer sensitivity [@yuan2023asvd].

SVD-LLM (https://arxiv.org/abs/2403.07378) combines truncation-aware whitening with sequential updates after compression [@wang2024svdllm]. Our fixed-rank diagonal ASVD and whitening-only SVD-LLM isolate their weighting components under a matched factor budget. They are not full-pipeline reproductions.

SVD-LLM V2 (https://aclanthology.org/2025.naacl-long.217/) changes rank allocation and singular-value truncation [@wang2025svdllmv2], so it belongs in a later adaptive-budget study rather than a claim that fixed per-matrix ranks reproduce its complete procedure.

Global composition is already an explicit target of related work.

SAES-SVD (https://arxiv.org/abs/2602.03051) addresses accumulated and local errors [@hu2026saessvd]; LACE-SVD (https://arxiv.org/abs/2607.03057) combines loss-aware allocation with residual-stream correction [@liu2026lacesvd].

Our smaller-group and downstream-drift comparisons test the location and extent

of composition damage, without claiming to resolve it. The next comparator is SAES-SVD if cumulative drift remains the decisive failure.

Quantization Error Propagation (https://arxiv.org/abs/2504.09629) demonstrates why accumulated errors also matter in layer-wise quantization [@arai2025qep]. That connection motivates measuring downstream behavior, but does not equate low-rank truncation with quantization or transfer an existing method's results.

Loss-aware global allocation provides further distinctions from this study's fixed-rank setup. Zero Sum SVD (https://arxiv.org/abs/2602.02848) uses whitened component sensitivities under a global pruning rule [@abbasi2026zerosum]. IO-SVD (https://arxiv.org/abs/2605.15626) adds an output-side predictive metric and heterogeneous rank selection [@abbasi2026iosvd]. These motivate future equal-total-budget comparisons. They do not establish novelty for an anchored input-covariance truncation or for the current extra-ridge solver.

The practical comparison distinguishes storage, accuracy, and execution cost. GPTQ (https://arxiv.org/abs/2210.17323) uses approximate second-order information for post-training quantization [@frantar2022gptq].

AWQ (https://arxiv.org/abs/2306.00978) uses activation-informed channel protection and an inference implementation [@lin2023awq].

SparseGPT (https://arxiv.org/abs/2301.00774) studies one-shot pruning [@frantar2023sparsegpt]. These are different representations, not matched fixed-rank baselines in the first wave. Their practical emphasis motivates recording actual checkpoint bytes and warmed end-to-end timings.

EoRA (https://arxiv.org/abs/2410.21271) compensates compressed models with eigenspace-informed low-rank branches [@liu2024eora].

SVDQuant (https://arxiv.org/abs/2411.05007) absorbs outliers in a high-precision low-rank branch while quantizing a residual branch, and identifies the runtime overhead of naive separate branch execution [@li2024svdquant]. SVDQuant's published scope is diffusion models. It supplies a system-design warning for future hybrid experiments here, not evidence of LLM quality or speedups.

CKA (https://proceedings.mlr.press/v97/kornblith19a.html) measures alignment between representation similarities [@kornblith2019cka]. We use it as a secondary diagnostic beside the predeclared normalized residual squared error; neither representation similarity nor norm preservation substitutes for held-out NLL or logit KL. Temperature scaling (https://proceedings.mlr.press/v70/guo17a.html) is a validation-fitted calibration control [@guo2017calibration], adapted here to next-token probabilities. Improvement after temperature fitting would not by itself identify an LLM denoising mechanism.

Evaluation conventions can change apparent model rankings; the reproducibility guidance in Biderman et al. (https://arxiv.org/abs/2405.14782) motivates pinned assets, task formats and declared target masking [@biderman2024evaluation].

Pointer Sentinel Mixture Models (https://arxiv.org/abs/1609.07843) introduced

WikiText [@merity2016wikitext]. Because WikiText has already informed this project, its present test evaluation is regression evidence. The later fresh confirmation samples a locked C4 shard, whose provenance is the T5 study (https://jmlr.org/papers/v21/20-074.html) [@raffel2020t5]. Neither its sampled confirmation nor the initial WikiText wave represents all text domains. HellaSwag (https://aclanthology.org/P19-1472/) supplies a commonsense completion benchmark [@zellers2019hellaswag], and ARC (https://arxiv.org/abs/1803.05457) supplies science-question reasoning with an explicit Challenge subset [@clark2018arc]. Those downstream evaluations remain pending; no downstream accuracy claim follows from the readiness controls.

# Recorded numerical controls

FP16 SVD readiness remains failed. Later BF16/FP8 variants are separate numerical cohorts, not repaired FP16 evidence. FP8 codes are decoded before BF16 multiplication; native tiny-GEMM support does not establish native model speed.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 2; match by Row ID.

| Row ID | Run / state | Variant | NLL | PPL |
|---|---|---|---|---|
| R001 | precision-readiness-20261014-060840-6db37daf / cancelled | teacher_bf16 | 3.4515495 | 31.5492399 |
| R002 | precision-readiness-20261014-060840-6db37daf / cancelled | copy_bf16 | 3.4515495 | 31.5492399 |
| R003 | precision-readiness-20261014-060840-6db37daf / cancelled | identity_bf16 | 3.4515495 | 31.5492399 |
| R004 | precision-readiness-20261014-060840-6db37daf / cancelled | svd_bf16 | 3.45265812 | 31.5842357 |
| R005 | precision-readiness-20261014-060840-6db37daf / cancelled | svd_fp32_compute | 3.45158372 | 31.5503197 |
| R006 | precision-readiness-20261014-060840-6db37daf / cancelled | svd_fp8_e4m3 | 3.45542964 | 31.6718934 |
| R007 | precision-readiness-20261014-060840-6db37daf / cancelled | svd_fp8_e5m2 | 3.47055948 | 32.1547273 |
| R008 | readiness-20261013-164122-cdc919e9 / failed | teacher | 3.45156254 | 31.5496515 |
| R009 | readiness-20261013-164122-cdc919e9 / failed | copied_teacher | 3.45156254 | 31.5496515 |
| R010 | readiness-20261013-164407-bda5b4d9 / completed | teacher | 3.45156254 | 31.5496515 |
| R011 | readiness-20261013-164407-bda5b4d9 / completed | copied_teacher | 3.45156254 | 31.5496515 |
| R012 | readiness-20261013-164407-bda5b4d9 / completed | full_rank_factorized | 3.45156254 | 31.5496515 |
| R013 | svd-diagnosis-20261013-165146-c3671715 / completed | teacher | 3.45156254 | 31.5496515 |
| R014 | svd-diagnosis-20261013-165146-c3671715 / completed | dense_svd_fp32 | 3.4516607 | 31.5527485 |
| R015 | svd-diagnosis-20261013-165146-c3671715 / completed | dense_svd_fp16 | 3.45170481 | 31.5541402 |
| R016 | svd-diagnosis-20261013-165146-c3671715 / completed | unbalanced_fp16 | 3.45186016 | 31.5590426 |
| R017 | svd-diagnosis-20261013-165146-c3671715 / completed | balanced_fp16 | 3.45184835 | 31.5586697 |
| R018 | svd-diagnosis-20261013-165146-c3671715 / completed | float32_execution | 3.4517713 | 31.5562382 |

Panel 2 of 2; match by Row ID.

| Row ID | Run / state | Maximum probe error | Probe gate | NLL gate |
|---|---|---|---|---|
| R001 | precision-readiness-20261014-060840-6db37daf / cancelled | 0 | True | True |
| R002 | precision-readiness-20261014-060840-6db37daf / cancelled | 0 | True | True |
| R003 | precision-readiness-20261014-060840-6db37daf / cancelled | 0 | True | True |
| R004 | precision-readiness-20261014-060840-6db37daf / cancelled | 1.0625 | False | False |
| R005 | precision-readiness-20261014-060840-6db37daf / cancelled | 1.15625 | False | True |
| R006 | precision-readiness-20261014-060840-6db37daf / cancelled | 9.73046875 | False | False |
| R007 | precision-readiness-20261014-060840-6db37daf / cancelled | 7.8671875 | False | False |
| R008 | readiness-20261013-164122-cdc919e9 / failed | 0 | True | unrecorded |
| R009 | readiness-20261013-164122-cdc919e9 / failed | 0 | True | unrecorded |
| R010 | readiness-20261013-164407-bda5b4d9 / completed | 0 | True | unrecorded |
| R011 | readiness-20261013-164407-bda5b4d9 / completed | 0 | True | unrecorded |
| R012 | readiness-20261013-164407-bda5b4d9 / completed | 0 | True | unrecorded |
| R013 | svd-diagnosis-20261013-165146-c3671715 / completed | unmeasured | unrecorded | unrecorded |
| R014 | svd-diagnosis-20261013-165146-c3671715 / completed | 0.1171875 | True | True |
| R015 | svd-diagnosis-20261013-165146-c3671715 / completed | 0.140625 | False | True |
| R016 | svd-diagnosis-20261013-165146-c3671715 / completed | 0.15234375 | False | True |
| R017 | svd-diagnosis-20261013-165146-c3671715 / completed | 0.181640625 | False | True |
| R018 | svd-diagnosis-20261013-165146-c3671715 / completed | 0.1171875 | True | True |

# Validation regularization controls

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R001 | regularization-20261013-171941-804f4c2e / completed | {"method": "svd"} | {} | [11] |
| R002 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.1, "method": "fullrank_ridge", "ridge_fraction": 0.01} | {} | [11] |
| R003 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {} | [11] |
| R004 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {} | [11] |

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R005 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.1, "method": "anchored", "ridge_fraction": 0.0} | {} | [11] |
| R006 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.3, "method": "anchored", "ridge_fraction": 0.0} | {} | [11] |
| R007 | regularization-20261013-171941-804f4c2e / completed | {"beta": 1.0, "method": "anchored", "ridge_fraction": 0.0} | {} | [11] |
| R008 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.0} | {} | [11] |
| R009 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.001} | {} | [11] |
| R010 | regularization-20261013-171941-804f4c2e / completed | {"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01} | {} | [11] |
| R011 | regularization-20261013-171941-804f4c2e / completed | {"method": "asvd"} | {} | [11] |

Panel 2 of 3; match by Row ID.

| Row ID | Run / state | NLL mean | NLL seed SD | Drift mean |
|---|---|---|---|---|
| R001 | regularization-20261013-171941-804f4c2e / completed | 5.14486078 | unmeasured | unmeasured |
| R002 | regularization-20261013-171941-804f4c2e / completed | 3.49923448 | unmeasured | unmeasured |
| R003 | regularization-20261013-171941-804f4c2e / completed | 3.60078899 | unmeasured | unmeasured |
| R004 | regularization-20261013-171941-804f4c2e / completed | 3.59662239 | unmeasured | unmeasured |
| R005 | regularization-20261013-171941-804f4c2e / completed | 3.59675176 | unmeasured | unmeasured |
| R006 | regularization-20261013-171941-804f4c2e / completed | 3.607465 | unmeasured | unmeasured |
| R007 | regularization-20261013-171941-804f4c2e / completed | 3.61579886 | unmeasured | unmeasured |
| R008 | regularization-20261013-171941-804f4c2e / completed | 3.59685438 | unmeasured | unmeasured |
| R009 | regularization-20261013-171941-804f4c2e / completed | 3.59727845 | unmeasured | unmeasured |
| R010 | regularization-20261013-171941-804f4c2e / completed | 3.60682971 | unmeasured | unmeasured |
| R011 | regularization-20261013-171941-804f4c2e / completed | 3.73607467 | unmeasured | unmeasured |

Panel 3 of 3; match by Row ID.

| Row ID | Run / state | Drift seed SD |
|---|---|---|
| R001 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R002 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R003 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R004 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R005 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R006 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R007 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R008 | regularization-20261013-171941-804f4c2e / completed | unmeasured |

| Row ID | Run / state | Drift seed SD |
|---|---|---|
| R009 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R010 | regularization-20261013-171941-804f4c2e / completed | unmeasured |
| R011 | regularization-20261013-171941-804f4c2e / completed | unmeasured |

# Gemma full-stack component comparisons

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R001 | baselines-20261013-175255-6a00d4e8 / completed | {"method": "svd"} | {} | [11] |
| R002 | baselines-20261013-175255-6a00d4e8 / completed | {"method": "asvd"} | {} | [11, 23, 37] |
| R003 | baselines-20261013-175255-6a00d4e8 / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {} | [11, 23, 37] |
| R004 | baselines-20261013-175255-6a00d4e8 / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {} | [11, 23, 37] |
| R005 | baselines-20261013-175255-6a00d4e8 / completed | {"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01} | {} | [11, 23, 37] |

Panel 2 of 3; match by Row ID.

| Row ID | Run / state | NLL mean | NLL seed SD | Drift mean |
|---|---|---|---|---|
| R001 | baselines-20261013-175255-6a00d4e8 / completed | 5.08868112 | unmeasured | 0.0742475831 |
| R002 | baselines-20261013-175255-6a00d4e8 / completed | 3.69446606 | 0.00241956806 | 0.0309717453 |
| R003 | baselines-20261013-175255-6a00d4e8 / completed | 3.57281383 | 0.0118191598 | 0.0106375166 |
| R004 | baselines-20261013-175255-6a00d4e8 / completed | 3.56592113 | 0.00957592299 | 0.0105470011 |
| R005 | baselines-20261013-175255-6a00d4e8 / completed | 3.57903427 | 0.00852077408 | 0.0118359709 |

Panel 3 of 3; match by Row ID.

| Row ID | Run / state | Drift seed SD |
|---|---|---|
| R001 | baselines-20261013-175255-6a00d4e8 / completed | unmeasured |
| R002 | baselines-20261013-175255-6a00d4e8 / completed | 0.00019731531 |
| R003 | baselines-20261013-175255-6a00d4e8 / completed | 0.000382136395 |
| R004 | baselines-20261013-175255-6a00d4e8 / completed | 0.000492296849 |
| R005 | baselines-20261013-175255-6a00d4e8 / completed | 0.000138552104 |

# Composition and group heterogeneity

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R001 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0], "rank": 384, "split": "test"} | [11, 23, 37] |
| R002 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0], "rank": 384, "split": "test"} | [11, 23, 37] |
| R003 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [12], "rank": 384, "split": "test"} | [11, 23, 37] |
| R004 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [12], "rank": 384, "split": "test"} | [11, 23, 37] |
| R005 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R006 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R007 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0, 1], "rank": 384, "split": "test"} | [11, 23, 37] |
| R008 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0, 1], "rank": 384, "split": "test"} | [11, 23, 37] |
| R009 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [12, 13], "rank": 384, "split": "test"} | [11, 23, 37] |
| R010 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [12, 13], "rank": 384, "split": "test"} | [11, 23, 37] |
| R011 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R012 | composition-20261013-185512-2bffd477 / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [24, 25], "rank": 384, "split": "test"} | [11] |
| R013 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0], "rank": 384, "split": "test"} | [11, 23, 37] |
| R014 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0], "rank": 384, "split": "test"} | [11, 23, 37] |
| R015 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [12], "rank": 384, "split": "test"} | [11, 23, 37] |
| R016 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [12], "rank": 384, "split": "test"} | [11, 23, 37] |
| R017 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R018 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R019 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0, 1], "rank": 384, "split": "test"} | [11, 23, 37] |
| R020 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0, 1], "rank": 384, "split": "test"} | [11, 23, 37] |
| R021 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [12, 13], "rank": 384, "split": "test"} | [11, 23, 37] |

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R022 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [12, 13], "rank": 384, "split": "test"} | [11, 23, 37] |
| R023 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R024 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R025 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0, 1, 2, 3], "rank": 384, "split": "test"} | [11, 23, 37] |
| R026 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0, 1, 2, 3], "rank": 384, "split": "test"} | [11, 23, 37] |
| R027 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [11, 12, 13, 14], "rank": 384, "split": "test"} | [11, 23, 37] |
| R028 | composition-20261014-032120-486d799f / interrupted | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [11, 12, 13, 14], "rank": 384, "split": "test"} | [11, 23] |
| R029 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0], "rank": 384, "split": "test"} | [11, 23, 37] |
| R030 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0], "rank": 384, "split": "test"} | [11, 23, 37] |
| R031 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [12], "rank": 384, "split": "test"} | [11, 23, 37] |
| R032 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [12], "rank": 384, "split": "test"} | [11, 23, 37] |
| R033 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R034 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R035 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0, 1], "rank": 384, "split": "test"} | [11, 23, 37] |
| R036 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0, 1], "rank": 384, "split": "test"} | [11, 23, 37] |
| R037 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [12, 13], "rank": 384, "split": "test"} | [11, 23, 37] |
| R038 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [12, 13], "rank": 384, "split": "test"} | [11, 23, 37] |
| R039 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R040 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R041 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0, 1, 2, 3], "rank": 384, "split": "test"} | [11, 23, 37] |
| R042 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0, 1, 2, 3], "rank": 384, "split": "test"} | [11, 23, 37] |

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R043 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [11, 12, 13, 14], "rank": 384, "split": "test"} | [11, 23, 37] |
| R044 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [11, 12, 13, 14], "rank": 384, "split": "test"} | [11, 23, 37] |
| R045 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [22, 23, 24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R046 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [22, 23, 24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R047 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |
| R048 | composition-20261014-054423-dbd821ff / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"} | [11, 23, 37] |

Panel 2 of 3; match by Row ID.

| Row ID | Run / state | NLL mean | NLL seed SD | Drift mean |
|---|---|---|---|---|
| R001 | composition-20261013-185512-2bffd477 / interrupted | 3.45206785 | 0.000410778252 | 0.00207792164 |
| R002 | composition-20261013-185512-2bffd477 / interrupted | 3.45201942 | 0.000409256997 | 0.00207452887 |
| R003 | composition-20261013-185512-2bffd477 / interrupted | 3.45825421 | 0.00132397251 | 8.76745161e-05 |
| R004 | composition-20261013-185512-2bffd477 / interrupted | 3.4579111 | 0.00143757277 | 8.91802058e-05 |
| R005 | composition-20261013-185512-2bffd477 / interrupted | 3.45142891 | 0.000329325004 | 1.69831209e-06 |
| R006 | composition-20261013-185512-2bffd477 / interrupted | 3.45122641 | 0.000131137763 | 2.023157e-06 |
| R007 | composition-20261013-185512-2bffd477 / interrupted | 3.45442908 | 0.00205462111 | 0.00365644585 |
| R008 | composition-20261013-185512-2bffd477 / interrupted | 3.45438131 | 0.00214728679 | 0.00356760921 |
| R009 | composition-20261013-185512-2bffd477 / interrupted | 3.46353003 | 0.00115146509 | 0.000141608504 |
| R010 | composition-20261013-185512-2bffd477 / interrupted | 3.46353582 | 0.00129669463 | 0.000144554107 |
| R011 | composition-20261013-185512-2bffd477 / interrupted | 3.45212529 | 0.000451146203 | 1.23266207e-05 |
| R012 | composition-20261013-185512-2bffd477 / interrupted | 3.45195513 | unmeasured | 1.21042136e-05 |
| R013 | composition-20261014-032120-486d799f / interrupted | 3.45206785 | 0.000410778252 | 0.00207792164 |
| R014 | composition-20261014-032120-486d799f / interrupted | 3.45201942 | 0.000409256997 | 0.00207452887 |
| R015 | composition-20261014-032120-486d799f / interrupted | 3.45825421 | 0.00132397251 | 8.76745161e-05 |
| R016 | composition-20261014-032120-486d799f / interrupted | 3.4579111 | 0.00143757277 | 8.91802058e-05 |
| R017 | composition-20261014-032120-486d799f / interrupted | 3.45142891 | 0.000329325004 | 1.69831209e-06 |
| R018 | composition-20261014-032120-486d799f / interrupted | 3.45122641 | 0.000131137763 | 2.023157e-06 |

| Row ID | Run / state | NLL mean | NLL seed SD | Drift mean |
|---|---|---|---|---|
| R019 | composition-20261014-032120-486d799f / interrupted | 3.45442908 | 0.00205462111 | 0.00365644585 |
| R020 | composition-20261014-032120-486d799f / interrupted | 3.45438131 | 0.00214728679 | 0.00356760921 |
| R021 | composition-20261014-032120-486d799f / interrupted | 3.46353003 | 0.00115146509 | 0.000141608504 |
| R022 | composition-20261014-032120-486d799f / interrupted | 3.46353582 | 0.00129669463 | 0.000144554107 |
| R023 | composition-20261014-032120-486d799f / interrupted | 3.45212529 | 0.000451146203 | 1.23266207e-05 |
| R024 | composition-20261014-032120-486d799f / interrupted | 3.45194198 | 0.000101102007 | 1.22970593e-05 |
| R025 | composition-20261014-032120-486d799f / interrupted | 3.46224947 | 0.000734496385 | 0.0061389891 |
| R026 | composition-20261014-032120-486d799f / interrupted | 3.4619553 | 0.00102918121 | 0.00600512161 |
| R027 | composition-20261014-032120-486d799f / interrupted | 3.47310172 | 0.000470679639 | 0.00038058521 |
| R028 | composition-20261014-032120-486d799f / interrupted | 3.4765793 | 1.58332576e-05 | 0.000438405635 |
| R029 | composition-20261014-054423-dbd821ff / completed | 3.45206785 | 0.000410778252 | 0.00207792164 |
| R030 | composition-20261014-054423-dbd821ff / completed | 3.45201942 | 0.000409256997 | 0.00207452887 |
| R031 | composition-20261014-054423-dbd821ff / completed | 3.45825421 | 0.00132397251 | 8.76745161e-05 |
| R032 | composition-20261014-054423-dbd821ff / completed | 3.4579111 | 0.00143757277 | 8.91802058e-05 |
| R033 | composition-20261014-054423-dbd821ff / completed | 3.45142891 | 0.000329325004 | 1.69831209e-06 |
| R034 | composition-20261014-054423-dbd821ff / completed | 3.45122641 | 0.000131137763 | 2.023157e-06 |
| R035 | composition-20261014-054423-dbd821ff / completed | 3.45442908 | 0.00205462111 | 0.00365644585 |
| R036 | composition-20261014-054423-dbd821ff / completed | 3.45438131 | 0.00214728679 | 0.00356760921 |
| R037 | composition-20261014-054423-dbd821ff / completed | 3.46353003 | 0.00115146509 | 0.000141608504 |
| R038 | composition-20261014-054423-dbd821ff / completed | 3.46353582 | 0.00129669463 | 0.000144554107 |
| R039 | composition-20261014-054423-dbd821ff / completed | 3.45212529 | 0.000451146203 | 1.23266207e-05 |
| R040 | composition-20261014-054423-dbd821ff / completed | 3.45194198 | 0.000101102007 | 1.22970593e-05 |
| R041 | composition-20261014-054423-dbd821ff / completed | 3.46224947 | 0.000734496385 | 0.0061389891 |
| R042 | composition-20261014-054423-dbd821ff / completed | 3.4619553 | 0.00102918121 | 0.00600512161 |
| R043 | composition-20261014-054423-dbd821ff / completed | 3.47310172 | 0.000470679639 | 0.00038058521 |
| R044 | composition-20261014-054423-dbd821ff / completed | 3.47582544 | 0.00130576482 | 0.000449140822 |
| R045 | composition-20261014-054423-dbd821ff / completed | 3.45396348 | 0.00136122732 | 0.000123680895 |
| R046 | composition-20261014-054423-dbd821ff / completed | 3.4529696 | 0.000385430754 | 0.00011315281 |
| R047 | composition-20261014-054423-dbd821ff / completed | 3.57281383 | 0.0118191598 | 0.0106375166 |

| Row ID | Run / state | NLL mean | NLL seed SD | Drift mean |
|---|---|---|---|---|
| R048 | composition-20261014-054423-dbd821ff / completed | 3.56592113 | 0.00957592299 | 0.0105470011 |

Panel 3 of 3; match by Row ID.

| Row ID | Run / state | Drift seed SD |
|---|---|---|
| R001 | composition-20261013-185512-2bffd477 / interrupted | 0.00017472334 |
| R002 | composition-20261013-185512-2bffd477 / interrupted | 0.000178507413 |
| R003 | composition-20261013-185512-2bffd477 / interrupted | 2.4463997e-06 |
| R004 | composition-20261013-185512-2bffd477 / interrupted | 2.24916162e-06 |
| R005 | composition-20261013-185512-2bffd477 / interrupted | 9.05406025e-08 |
| R006 | composition-20261013-185512-2bffd477 / interrupted | 9.42299447e-08 |
| R007 | composition-20261013-185512-2bffd477 / interrupted | 0.000137627131 |
| R008 | composition-20261013-185512-2bffd477 / interrupted | 0.000238304358 |
| R009 | composition-20261013-185512-2bffd477 / interrupted | 4.06719991e-06 |
| R010 | composition-20261013-185512-2bffd477 / interrupted | 4.09979657e-06 |
| R011 | composition-20261013-185512-2bffd477 / interrupted | 3.68456941e-07 |
| R012 | composition-20261013-185512-2bffd477 / interrupted | unmeasured |
| R013 | composition-20261014-032120-486d799f / interrupted | 0.00017472334 |
| R014 | composition-20261014-032120-486d799f / interrupted | 0.000178507413 |
| R015 | composition-20261014-032120-486d799f / interrupted | 2.4463997e-06 |
| R016 | composition-20261014-032120-486d799f / interrupted | 2.24916162e-06 |
| R017 | composition-20261014-032120-486d799f / interrupted | 9.05406025e-08 |
| R018 | composition-20261014-032120-486d799f / interrupted | 9.42299447e-08 |
| R019 | composition-20261014-032120-486d799f / interrupted | 0.000137627131 |
| R020 | composition-20261014-032120-486d799f / interrupted | 0.000238304358 |
| R021 | composition-20261014-032120-486d799f / interrupted | 4.06719991e-06 |
| R022 | composition-20261014-032120-486d799f / interrupted | 4.09979657e-06 |
| R023 | composition-20261014-032120-486d799f / interrupted | 3.68456941e-07 |
| R024 | composition-20261014-032120-486d799f / interrupted | 3.91576035e-07 |
| R025 | composition-20261014-032120-486d799f / interrupted | 0.000142663396 |
| R026 | composition-20261014-032120-486d799f / interrupted | 0.000161076046 |
| R027 | composition-20261014-032120-486d799f / interrupted | 1.72440575e-05 |
| R028 | composition-20261014-032120-486d799f / interrupted | 3.31525237e-07 |
| R029 | composition-20261014-054423-dbd821ff / completed | 0.00017472334 |
| R030 | composition-20261014-054423-dbd821ff / completed | 0.000178507413 |
| R031 | composition-20261014-054423-dbd821ff / completed | 2.4463997e-06 |
| R032 | composition-20261014-054423-dbd821ff / completed | 2.24916162e-06 |
| R033 | composition-20261014-054423-dbd821ff / completed | 9.05406025e-08 |
| R034 | composition-20261014-054423-dbd821ff / completed | 9.42299447e-08 |
| R035 | composition-20261014-054423-dbd821ff / completed | 0.000137627131 |
| R036 | composition-20261014-054423-dbd821ff / completed | 0.000238304358 |
| R037 | composition-20261014-054423-dbd821ff / completed | 4.06719991e-06 |
| R038 | composition-20261014-054423-dbd821ff / completed | 4.09979657e-06 |
| R039 | composition-20261014-054423-dbd821ff / completed | 3.68456941e-07 |

| Row ID | Run / state | Drift seed SD |
|---|---|---|
| R040 | composition-20261014-054423-dbd821ff / completed | 3.91576035e-07 |
| R041 | composition-20261014-054423-dbd821ff / completed | 0.000142663396 |
| R042 | composition-20261014-054423-dbd821ff / completed | 0.000161076046 |
| R043 | composition-20261014-054423-dbd821ff / completed | 1.72440575e-05 |
| R044 | composition-20261014-054423-dbd821ff / completed | 1.85953677e-05 |
| R045 | composition-20261014-054423-dbd821ff / completed | 1.21816693e-06 |
| R046 | composition-20261014-054423-dbd821ff / completed | 2.34008283e-06 |
| R047 | composition-20261014-054423-dbd821ff / completed | 0.000382136395 |
| R048 | composition-20261014-054423-dbd821ff / completed | 0.000492296849 |

Group-dependent outcomes must not be pooled into a universal-improvement claim. A lower full-stack NLL with unchanged/worse residual drift does not satisfy the predeclared joint gate. Three paired seeds with both higher NLL and higher drift weaken the candidate only at that measured scope.

# Amended hybrid architecture and rank-frontier comparisons

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R001 | architecture-replication-20261014-093845-97378b5e / incomplete | {"method": "teacher"} | {"rank": null, "rank_divisor": null, "recipe_ranks": [null, null, null, null, null, null], "split": "test"} | [11] |
| R002 | architecture-replication-20261014-093845-97378b5e / incomplete | {"method": "svd"} | {"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"} | [11] |
| R003 | architecture-replication-20261014-093845-97378b5e / incomplete | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"} | [11, 23, 37] |
| R004 | architecture-replication-20261014-093845-97378b5e / incomplete | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"} | [11, 23, 37] |
| R005 | architecture-replication-20261014-093845-97378b5e / incomplete | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"rank": 512, "rank_divisor": 2, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"} | [11, 23, 37] |
| R006 | architecture-replication-20261014-093845-97378b5e / incomplete | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"rank": 512, "rank_divisor": 2, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"} | [11, 23, 37] |
| R007 | architecture-replication-20261014-093845-97378b5e / incomplete | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"rank": 341, "rank_divisor": 3, "recipe_ranks": [341, 341, 341, 341, 341, 341], "split": "validation"} | [11, 23] |
| R008 | architecture-replication-20261014-093845-97378b5e / incomplete | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"rank": 341, "rank_divisor": 3, "recipe_ranks": [341, 341, 341, 341, 341, 341], "split": "validation"} | [11, 23] |
| R009 | architecture-replication-20261014-103759-8d32ab84 / completed | {"method": "teacher"} | {"rank": null, "rank_divisor": null, "recipe_ranks": [null, null, null, null, null, null], "split": "test"} | [11] |
| R010 | architecture-replication-20261014-103759-8d32ab84 / completed | {"method": "svd"} | {"rank": 725, "rank_divisor": null, "recipe_ranks": [725, 725, 725, 725, 725, 725], "split": "test"} | [11] |

| Row ID | Run / state | Specification | Split / group / rank scope | Seeds |
|---|---|---|---|---|
| R011 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"rank": 725, "rank_divisor": null, "recipe_ranks": [725, 725, 725, 725, 725, 725], "split": "test"} | [11, 23, 37] |
| R012 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"rank": 725, "rank_divisor": null, "recipe_ranks": [725, 725, 725, 725, 725, 725], "split": "test"} | [11, 23, 37] |
| R013 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"rank": 1024, "rank_divisor": 2, "recipe_ranks": [1024, 1024, 1024, 1024, 1024, 1024], "split": "validation"} | [11, 23, 37] |
| R014 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"rank": 1024, "rank_divisor": 2, "recipe_ranks": [1024, 1024, 1024, 1024, 1024, 1024], "split": "validation"} | [11, 23, 37] |
| R015 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"rank": 682, "rank_divisor": 3, "recipe_ranks": [682, 682, 682, 682, 682, 682], "split": "validation"} | [11, 23, 37] |
| R016 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"rank": 682, "rank_divisor": 3, "recipe_ranks": [682, 682, 682, 682, 682, 682], "split": "validation"} | [11, 23, 37] |
| R017 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0} | {"rank": 512, "rank_divisor": 4, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"} | [11, 23, 37] |
| R018 | architecture-replication-20261014-103759-8d32ab84 / completed | {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0} | {"rank": 512, "rank_divisor": 4, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"} | [11, 23, 37] |

Panel 2 of 3; match by Row ID.

| Row ID | Run / state | NLL mean | NLL seed SD | Drift mean |
|---|---|---|---|---|
| R001 | architecture-replication-20261014-093845-97378b5e / incomplete | 2.91722043 | unmeasured | 0 |
| R002 | architecture-replication-20261014-093845-97378b5e / incomplete | 3.18099289 | unmeasured | 0.0616629847 |
| R003 | architecture-replication-20261014-093845-97378b5e / incomplete | 2.93443868 | 0.00145544439 | 0.00615478758 |
| R004 | architecture-replication-20261014-093845-97378b5e / incomplete | 2.94332386 | 0.00227178531 | 0.0127619533 |
| R005 | architecture-replication-20261014-093845-97378b5e / incomplete | 2.98576098 | 0.00186708519 | 0.00527501643 |
| R006 | architecture-replication-20261014-093845-97378b5e / incomplete | 2.99344161 | 0.00188966622 | 0.0110389616 |
| R007 | architecture-replication-20261014-093845-97378b5e / incomplete | 3.00960245 | 0.00513867909 | 0.0111782273 |
| R008 | architecture-replication-20261014-093845-97378b5e / incomplete | 3.03686124 | 0.00451877599 | 0.0223829381 |
| R009 | architecture-replication-20261014-103759-8d32ab84 / completed | 3.99873556 | unmeasured | 0 |
| R010 | architecture-replication-20261014-103759-8d32ab84 / completed | 3.85243198 | unmeasured | 0.237778731 |

| Row ID | Run / state | NLL mean | NLL seed SD | Drift mean |
|---|---|---|---|---|
| R011 | architecture-replication-20261014-103759-8d32ab84 / completed | 4.05640463 | 0.0228864502 | 0.0358803942 |
| R012 | architecture-replication-20261014-103759-8d32ab84 / completed | 3.83591612 | 0.0103346354 | 0.13834591 |
| R013 | architecture-replication-20261014-103759-8d32ab84 / completed | 4.11730048 | 0.00999990552 | 0.0131921335 |
| R014 | architecture-replication-20261014-103759-8d32ab84 / completed | 3.99709851 | 0.0034649186 | 0.0500460576 |
| R015 | architecture-replication-20261014-103759-8d32ab84 / completed | 4.15837853 | 0.0193157214 | 0.0388193871 |
| R016 | architecture-replication-20261014-103759-8d32ab84 / completed | 3.88534704 | 0.00924738816 | 0.1558375 |
| R017 | architecture-replication-20261014-103759-8d32ab84 / completed | 4.22238934 | 0.0312371256 | 0.0634802611 |
| R018 | architecture-replication-20261014-103759-8d32ab84 / completed | 3.89287487 | 0.00650573852 | 0.247307618 |

Panel 3 of 3; match by Row ID.

| Row ID | Run / state | Drift seed SD |
|---|---|---|
| R001 | architecture-replication-20261014-093845-97378b5e / incomplete | unmeasured |
| R002 | architecture-replication-20261014-093845-97378b5e / incomplete | unmeasured |
| R003 | architecture-replication-20261014-093845-97378b5e / incomplete | 9.06818768e-05 |
| R004 | architecture-replication-20261014-093845-97378b5e / incomplete | 0.000160025705 |
| R005 | architecture-replication-20261014-093845-97378b5e / incomplete | 0.000155706933 |
| R006 | architecture-replication-20261014-093845-97378b5e / incomplete | 7.87701617e-05 |
| R007 | architecture-replication-20261014-093845-97378b5e / incomplete | 0.000317143432 |
| R008 | architecture-replication-20261014-093845-97378b5e / incomplete | 0.000308259687 |
| R009 | architecture-replication-20261014-103759-8d32ab84 / completed | unmeasured |
| R010 | architecture-replication-20261014-103759-8d32ab84 / completed | unmeasured |
| R011 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.000777615937 |
| R012 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.000864376837 |
| R013 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.000154870713 |
| R014 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.000338835218 |
| R015 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.000366967559 |
| R016 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.00119386663 |

| Row ID | Run / state | Drift seed SD |
|---|---|---|
| R017 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.000757422494 |
| R018 | architecture-replication-20261014-103759-8d32ab84 / completed | 0.00126797867 |

Recovery provenance: {"independent_replication": false, "interrupted_cost": {"basis": "Conservative actual Python process launch-to-next-system-boot wall interval; includes reboot gap; original status retained unchanged.", "charged_seconds": 3647.1586377620697, "independent_replication": false, "model": "Qwen/Qwen3.5-0.8B", "new_boot_epoch": 1791970132.8027894, "records_saved": 10, "run": "C:\\Users\\NPC\\Desktop\\ResearchCompression\\results\\paper-v2\\architecture-replication-20261014-082807-5c73f669", "run_id": "architecture-replication-20261014-082807-5c73f669", "source_artifacts": {"config.json": "d9a0760000b6e7222b9671a5e0561d61bb9f361355fe777766d5189dcead9898", "contract.json": "703af918c7287806089ad2ec358b9df623af24552ce7b580e0c6fd462a680ab9", "metrics.json": "29422a72a2536d51795bd97abfb112e2a53d2957bed9e0b8be0bcab49aa4c621", "readiness.json": "067bae33ea5bd31a127520d46deadc34a79504cd987a948fd052e7aca31c50f3", "status.json": "2f66cc74973ccbf05b72874c6500e8b742ee6ad8d80770d52b2b81f10d8c32fd", "validation-teacher.json": "057c49abf6c7728756cf3acd2af6f03163b416d9195be9df34fff95cb88aeeb8"}, "stage": "replication", "started_epoch": 1791966485.6441517, "state": "interrupted by reboot"}, "reused_records": 10, "source_artifacts": {"config.json": "d9a0760000b6e7222b9671a5e0561d61bb9f361355fe777766d5189dcead9898", "contract.json": "703af918c7287806089ad2ec358b9df623af24552ce7b580e0c6fd462a680ab9", "metrics.json": "29422a72a2536d51795bd97abfb112e2a53d2957bed9e0b8be0bcab49aa4c621", "readiness.json": "067bae33ea5bd31a127520d46deadc34a79504cd987a948fd052e7aca31c50f3", "status.json": "2f66cc74973ccbf05b72874c6500e8b742ee6ad8d80770d52b2b81f10d8c32fd", "validation-teacher.json": "057c49abf6c7728756cf3acd2af6f03163b416d9195be9df34fff95cb88aeeb8"}, "source_run": "C:\\Users\\NPC\\Desktop\\ResearchCompression\\results\\paper-v2\\architecture-replication-20261014-082807-5c73f669"}. Successor includes preserved source observations; attempts remain separate and are not independent replications.

Scoped finding (architecture-replication-20261014-093845-97378b5e, Qwen/Qwen3.5-0.8B): at the primary test rank, anchoring has higher NLL and higher primary drift than matched whitening in all three calibration seeds. This weakens the anchor hypothesis at this measured scope; it does not establish a universal failure or count reused observations as new replications.

Scoped tradeoff (architecture-replication-20261014-103759-8d32ab84, LiquidAI/LFM2.5-1.2B-Instruct): at the primary test rank, anchoring lowers NLL but increases primary drift versus matched whitening in every historical calibration seed. The likelihood improvement does not establish preserved hidden-state computation or a denoising mechanism. This mixed outcome remains inconclusive under the joint criterion.

# Practical costs and calibration

Whole-model checkpoint bytes include archive overhead; factor payload size and whole-model savings are separate quantities. Placements and precision must agree before comparing timings. Below are measured records only; missing conditions remain unmeasured.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 5; match by Row ID.

| Row ID | Run / state | Label | Seed | Bits |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 / cancelled | teacher | None | dense / float16 |
| R002 | practical-20261014-113642-0fe078b5 / cancelled | teacher | None | dense / float16 |
| R003 | practical-20261014-113642-0fe078b5 / cancelled | teacher | None | dense / float16 |
| R004 | practical-20261014-113642-0fe078b5 / cancelled | teacher | None | dense / float16 |
| R005 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 16 / float16 |
| R006 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 16 / float16 |
| R007 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 16 / float16 |

| Row ID | Run / state | Label | Seed | Bits |
|---|---|---|---|---|
| R008 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 16 / float16 |
| R009 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 16 / float16 |
| R010 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 16 / float16 |
| R011 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 16 / float16 |
| R012 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 16 / float16 |
| R013 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 8 / float16 |
| R014 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 8 / float16 |
| R015 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 8 / float16 |
| R016 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 11 | 8 / float16 |
| R017 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 8 / float16 |
| R018 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 8 / float16 |
| R019 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 8 / float16 |
| R020 | practical-20261014-113642-0fe078b5 / cancelled | baseline | 23 | 8 / float16 |
| R021 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 16 / float16 |
| R022 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 16 / float16 |
| R023 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 16 / float16 |
| R024 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 16 / float16 |
| R025 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 8 / float16 |
| R026 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 8 / float16 |
| R027 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 8 / float16 |
| R028 | practical-20261014-113642-0fe078b5 / cancelled | solver | 11 | 8 / float16 |
| R029 | practical-bf16-20261014-135107-82168ad5 / completed | teacher | None | 16 / bfloat16 |
| R030 | practical-bf16-20261014-135107-82168ad5 / completed | teacher | None | 16 / bfloat16 |
| R031 | practical-bf16-20261014-135107-82168ad5 / completed | teacher | None | 16 / bfloat16 |
| R032 | practical-bf16-20261014-135107-82168ad5 / completed | teacher | None | 16 / bfloat16 |
| R033 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 11 | 16 / bfloat16 |
| R034 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 11 | 16 / bfloat16 |
| R035 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 11 | 16 / bfloat16 |
| R036 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 11 | 16 / bfloat16 |

| Row ID | Run / state | Label | Seed | Bits |
|---|---|---|---|---|
| R037 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 23 | 16 / bfloat16 |
| R038 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 23 | 16 / bfloat16 |
| R039 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 23 | 16 / bfloat16 |
| R040 | practical-bf16-20261014-135107-82168ad5 / completed | baseline | 23 | 16 / bfloat16 |
| R041 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 11 | 16 / bfloat16 |
| R042 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 11 | 16 / bfloat16 |
| R043 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 11 | 16 / bfloat16 |
| R044 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 11 | 16 / bfloat16 |
| R045 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 23 | 16 / bfloat16 |
| R046 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 23 | 16 / bfloat16 |
| R047 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 23 | 16 / bfloat16 |
| R048 | practical-bf16-20261014-135107-82168ad5 / completed | solver | 23 | 16 / bfloat16 |

Panel 2 of 5; match by Row ID.

| Row ID | Run / state | Whole bytes | Factor bytes | Raw NLL |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 / cancelled | 1.99988434e+09 | unmeasured | 3.45156254 |
| R002 | practical-20261014-113642-0fe078b5 / cancelled | 1.99988434e+09 | unmeasured | 3.45156254 |
| R003 | practical-20261014-113642-0fe078b5 / cancelled | 1.99988434e+09 | unmeasured | 3.45156254 |
| R004 | practical-20261014-113642-0fe078b5 / cancelled | 1.99988434e+09 | unmeasured | 3.45156254 |
| R005 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56275409 |
| R006 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56275409 |
| R007 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56275409 |
| R008 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56275409 |
| R009 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56985661 |
| R010 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56985661 |
| R011 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56985661 |
| R012 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.56985661 |
| R013 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56308141 |
| R014 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56308141 |
| R015 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56308141 |

| Row ID | Run / state | Whole bytes | Factor bytes | Raw NLL |
|---|---|---|---|---|
| R016 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56308141 |
| R017 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56346292 |
| R018 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56346292 |
| R019 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56346292 |
| R020 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56346292 |
| R021 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.55963483 |
| R022 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.55963483 |
| R023 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.55963483 |
| R024 | practical-20261014-113642-0fe078b5 / cancelled | 1.98200268e+09 | 43509058 | 3.55963483 |
| R025 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56194517 |
| R026 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56194517 |
| R027 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56194517 |
| R028 | practical-20261014-113642-0fe078b5 / cancelled | 1.96061379e+09 | 22118386 | 3.56194517 |
| R029 | practical-bf16-20261014-135107-82168ad5 / completed | 1.99988371e+09 | unmeasured | 3.4515495 |
| R030 | practical-bf16-20261014-135107-82168ad5 / completed | 1.99988371e+09 | unmeasured | 3.4515495 |
| R031 | practical-bf16-20261014-135107-82168ad5 / completed | 1.99988371e+09 | unmeasured | 3.4515495 |
| R032 | practical-bf16-20261014-135107-82168ad5 / completed | 1.99988371e+09 | unmeasured | 3.4515495 |
| R033 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.55608618 |
| R034 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.55608618 |
| R035 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.55608618 |
| R036 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.55608618 |
| R037 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56843147 |
| R038 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56843147 |
| R039 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56843147 |
| R040 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56843147 |
| R041 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.5568263 |
| R042 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.5568263 |
| R043 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.5568263 |
| R044 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.5568263 |

| Row ID | Run / state | Whole bytes | Factor bytes | Raw NLL |
|---|---|---|---|---|
| R045 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56294882 |
| R046 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56294882 |
| R047 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56294882 |
| R048 | practical-bf16-20261014-135107-82168ad5 / completed | 1.98200351e+09 | 43467335 | 3.56294882 |

Panel 3 of 5; match by Row ID.

| Row ID | Run / state | Scaled NLL | Temperature | Mode |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 / cancelled | 3.42255866 | 1.09050773 | prefill |
| R002 | practical-20261014-113642-0fe078b5 / cancelled | 3.42255866 | 1.09050773 | decode |
| R003 | practical-20261014-113642-0fe078b5 / cancelled | 3.42255866 | 1.09050773 | prefill |
| R004 | practical-20261014-113642-0fe078b5 / cancelled | 3.42255866 | 1.09050773 | decode |
| R005 | practical-20261014-113642-0fe078b5 / cancelled | 3.54815343 | 1.09050773 | prefill |
| R006 | practical-20261014-113642-0fe078b5 / cancelled | 3.54815343 | 1.09050773 | decode |
| R007 | practical-20261014-113642-0fe078b5 / cancelled | 3.54815343 | 1.09050773 | prefill |
| R008 | practical-20261014-113642-0fe078b5 / cancelled | 3.54815343 | 1.09050773 | decode |
| R009 | practical-20261014-113642-0fe078b5 / cancelled | 3.55435184 | 1.09050773 | prefill |
| R010 | practical-20261014-113642-0fe078b5 / cancelled | 3.55435184 | 1.09050773 | decode |
| R011 | practical-20261014-113642-0fe078b5 / cancelled | 3.55435184 | 1.09050773 | prefill |
| R012 | practical-20261014-113642-0fe078b5 / cancelled | 3.55435184 | 1.09050773 | decode |
| R013 | practical-20261014-113642-0fe078b5 / cancelled | 3.54883071 | 1.09050773 | prefill |
| R014 | practical-20261014-113642-0fe078b5 / cancelled | 3.54883071 | 1.09050773 | decode |
| R015 | practical-20261014-113642-0fe078b5 / cancelled | 3.54883071 | 1.09050773 | prefill |
| R016 | practical-20261014-113642-0fe078b5 / cancelled | 3.54883071 | 1.09050773 | decode |
| R017 | practical-20261014-113642-0fe078b5 / cancelled | 3.54777084 | 1.09050773 | prefill |
| R018 | practical-20261014-113642-0fe078b5 / cancelled | 3.54777084 | 1.09050773 | decode |
| R019 | practical-20261014-113642-0fe078b5 / cancelled | 3.54777084 | 1.09050773 | prefill |
| R020 | practical-20261014-113642-0fe078b5 / cancelled | 3.54777084 | 1.09050773 | decode |
| R021 | practical-20261014-113642-0fe078b5 / cancelled | 3.54464574 | 1.09050773 | prefill |
| R022 | practical-20261014-113642-0fe078b5 / cancelled | 3.54464574 | 1.09050773 | decode |
| R023 | practical-20261014-113642-0fe078b5 / cancelled | 3.54464574 | 1.09050773 | prefill |

| Row ID | Run / state | Scaled NLL | Temperature | Mode |
|---|---|---|---|---|
| R024 | practical-20261014-113642-0fe078b5 / cancelled | 3.54464574 | 1.09050773 | decode |
| R025 | practical-20261014-113642-0fe078b5 / cancelled | 3.54677328 | 1.09050773 | prefill |
| R026 | practical-20261014-113642-0fe078b5 / cancelled | 3.54677328 | 1.09050773 | decode |
| R027 | practical-20261014-113642-0fe078b5 / cancelled | 3.54677328 | 1.09050773 | prefill |
| R028 | practical-20261014-113642-0fe078b5 / cancelled | 3.54677328 | 1.09050773 | decode |
| R029 | practical-bf16-20261014-135107-82168ad5 / completed | 3.42260056 | 1.09050773 | prefill |
| R030 | practical-bf16-20261014-135107-82168ad5 / completed | 3.42260056 | 1.09050773 | decode |
| R031 | practical-bf16-20261014-135107-82168ad5 / completed | 3.42260056 | 1.09050773 | prefill |
| R032 | practical-bf16-20261014-135107-82168ad5 / completed | 3.42260056 | 1.09050773 | decode |
| R033 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54172555 | 1.09050773 | prefill |
| R034 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54172555 | 1.09050773 | decode |
| R035 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54172555 | 1.09050773 | prefill |
| R036 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54172555 | 1.09050773 | decode |
| R037 | practical-bf16-20261014-135107-82168ad5 / completed | 3.55326688 | 1.09050773 | prefill |
| R038 | practical-bf16-20261014-135107-82168ad5 / completed | 3.55326688 | 1.09050773 | decode |
| R039 | practical-bf16-20261014-135107-82168ad5 / completed | 3.55326688 | 1.09050773 | prefill |
| R040 | practical-bf16-20261014-135107-82168ad5 / completed | 3.55326688 | 1.09050773 | decode |
| R041 | practical-bf16-20261014-135107-82168ad5 / completed | 3.5422546 | 1.09050773 | prefill |
| R042 | practical-bf16-20261014-135107-82168ad5 / completed | 3.5422546 | 1.09050773 | decode |
| R043 | practical-bf16-20261014-135107-82168ad5 / completed | 3.5422546 | 1.09050773 | prefill |
| R044 | practical-bf16-20261014-135107-82168ad5 / completed | 3.5422546 | 1.09050773 | decode |
| R045 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54534144 | 1.09050773 | prefill |
| R046 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54534144 | 1.09050773 | decode |
| R047 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54534144 | 1.09050773 | prefill |
| R048 | practical-bf16-20261014-135107-82168ad5 / completed | 3.54534144 | 1.09050773 | decode |

Panel 4 of 5; match by Row ID.

| Row ID | Run / state | Prefill | Median s | p95 s |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 0.05783475 | 0.059800255 |
| R002 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 6.4778003 | 6.64036795 |

| Row ID | Run / state | Prefill | Median s | p95 s |
|---|---|---|---|---|
| R003 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 0.055825 | 0.05969351 |
| R004 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 6.05921945 | 6.1403141 |
| R005 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 0.0590297 | 0.08538865 |
| R006 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 6.44498855 | 6.62319019 |
| R007 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 0.05369865 | 0.055011425 |
| R008 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 6.11049025 | 6.4726321 |
| R009 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 0.05889345 | 0.06917032 |
| R010 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 6.59023655 | 6.83514165 |
| R011 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 0.05657515 | 0.06035711 |
| R012 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 6.14512885 | 6.47288771 |
| R013 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 0.06479795 | 0.09071199 |
| R014 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 7.11784765 | 7.23628017 |
| R015 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 0.0592873 | 0.06291703 |
| R016 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 6.7235441 | 6.79397279 |
| R017 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 0.0648798 | 0.066804915 |
| R018 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 7.2303226 | 8.77872235 |
| R019 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 0.0611183 | 0.065548205 |
| R020 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 7.09512565 | 7.33472052 |
| R021 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 0.0603992 | 0.072276215 |
| R022 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 6.6212441 | 6.8661912 |
| R023 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 0.0581729 | 0.06183209 |
| R024 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 6.18178745 | 6.36525901 |
| R025 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 0.06678435 | 0.072467045 |
| R026 | practical-20261014-113642-0fe078b5 / cancelled | 128 | 7.19626915 | 7.34018282 |
| R027 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 0.06112195 | 0.064887285 |
| R028 | practical-20261014-113642-0fe078b5 / cancelled | 512 | 6.7455395 | 6.95685991 |
| R029 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 0.05917745 | 0.070606515 |
| R030 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 6.42761055 | 6.498819 |
| R031 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 0.0544619 | 0.057006585 |

| Row ID | Run / state | Prefill | Median s | p95 s |
|---|---|---|---|---|
| R032 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 6.23879075 | 6.43212847 |
| R033 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 0.0620393 | 0.072688375 |
| R034 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 6.65239975 | 6.93452725 |
| R035 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 0.0567413 | 0.06545248 |
| R036 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 6.3900002 | 6.60615606 |
| R037 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 0.06372255 | 0.07085632 |
| R038 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 6.6857301 | 7.01892626 |
| R039 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 0.05767275 | 0.05970958 |
| R040 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 6.40245935 | 6.74076871 |
| R041 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 0.07367855 | 0.08038677 |
| R042 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 7.36599745 | 8.0900931 |
| R043 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 0.0573537 | 0.0659666 |
| R044 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 6.31099465 | 6.43883127 |
| R045 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 0.05970755 | 0.06792561 |
| R046 | practical-bf16-20261014-135107-82168ad5 / completed | 128 | 6.4710207 | 6.57088731 |
| R047 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 0.05463395 | 0.056230945 |
| R048 | practical-bf16-20261014-135107-82168ad5 / completed | 512 | 6.1332589 | 6.25817783 |

Panel 5 of 5; match by Row ID.

| Row ID | Run / state | Allocated bytes | Reserved bytes |
|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 / cancelled | 2.13196698e+09 | 2.1495808e+09 |
| R002 | practical-20261014-113642-0fe078b5 / cancelled | 2.07219046e+09 | 2.15587226e+09 |
| R003 | practical-20261014-113642-0fe078b5 / cancelled | 2.34440499e+09 | 2.40543334e+09 |
| R004 | practical-20261014-113642-0fe078b5 / cancelled | 2.08504525e+09 | 2.40543334e+09 |
| R005 | practical-20261014-113642-0fe078b5 / cancelled | 2.1154519e+09 | 2.19571814e+09 |
| R006 | practical-20261014-113642-0fe078b5 / cancelled | 2.05567539e+09 | 2.1978153e+09 |
| R007 | practical-20261014-113642-0fe078b5 / cancelled | 2.32788992e+09 | 2.44527923e+09 |
| R008 | practical-20261014-113642-0fe078b5 / cancelled | 2.06859571e+09 | 2.44527923e+09 |
| R009 | practical-20261014-113642-0fe078b5 / cancelled | 2.1154519e+09 | 2.19571814e+09 |
| R010 | practical-20261014-113642-0fe078b5 / cancelled | 2.05567539e+09 | 2.1978153e+09 |

| Row ID | Run / state | Allocated bytes | Reserved bytes |
|---|---|---|---|
| R011 | practical-20261014-113642-0fe078b5 / cancelled | 2.32788992e+09 | 2.44527923e+09 |
| R012 | practical-20261014-113642-0fe078b5 / cancelled | 2.06859571e+09 | 2.44527923e+09 |
| R013 | practical-20261014-113642-0fe078b5 / cancelled | 2.09404621e+09 | 2.16426086e+09 |
| R014 | practical-20261014-113642-0fe078b5 / cancelled | 2.03845376e+09 | 2.16845517e+09 |
| R015 | practical-20261014-113642-0fe078b5 / cancelled | 2.30648422e+09 | 2.42011341e+09 |
| R016 | practical-20261014-113642-0fe078b5 / cancelled | 2.04719002e+09 | 2.42011341e+09 |
| R017 | practical-20261014-113642-0fe078b5 / cancelled | 2.09404621e+09 | 2.16426086e+09 |
| R018 | practical-20261014-113642-0fe078b5 / cancelled | 2.03845376e+09 | 2.16845517e+09 |
| R019 | practical-20261014-113642-0fe078b5 / cancelled | 2.30648422e+09 | 2.42011341e+09 |
| R020 | practical-20261014-113642-0fe078b5 / cancelled | 2.04719002e+09 | 2.42011341e+09 |
| R021 | practical-20261014-113642-0fe078b5 / cancelled | 2.1154519e+09 | 2.19571814e+09 |
| R022 | practical-20261014-113642-0fe078b5 / cancelled | 2.05567539e+09 | 2.1978153e+09 |
| R023 | practical-20261014-113642-0fe078b5 / cancelled | 2.32788992e+09 | 2.44527923e+09 |
| R024 | practical-20261014-113642-0fe078b5 / cancelled | 2.06859571e+09 | 2.44527923e+09 |
| R025 | practical-20261014-113642-0fe078b5 / cancelled | 2.09404621e+09 | 2.16426086e+09 |
| R026 | practical-20261014-113642-0fe078b5 / cancelled | 2.03845376e+09 | 2.16845517e+09 |
| R027 | practical-20261014-113642-0fe078b5 / cancelled | 2.30648422e+09 | 2.42011341e+09 |
| R028 | practical-20261014-113642-0fe078b5 / cancelled | 2.04719002e+09 | 2.42011341e+09 |
| R029 | practical-bf16-20261014-135107-82168ad5 / completed | 2.13196698e+09 | 2.1495808e+09 |
| R030 | practical-bf16-20261014-135107-82168ad5 / completed | 2.07219046e+09 | 2.15587226e+09 |
| R031 | practical-bf16-20261014-135107-82168ad5 / completed | 2.34440499e+09 | 2.40543334e+09 |
| R032 | practical-bf16-20261014-135107-82168ad5 / completed | 2.08504525e+09 | 2.40543334e+09 |
| R033 | practical-bf16-20261014-135107-82168ad5 / completed | 2.12226765e+09 | 2.1537751e+09 |
| R034 | practical-bf16-20261014-135107-82168ad5 / completed | 2.06249114e+09 | 2.15587226e+09 |
| R035 | practical-bf16-20261014-135107-82168ad5 / completed | 2.33470566e+09 | 2.38865613e+09 |
| R036 | practical-bf16-20261014-135107-82168ad5 / completed | 2.07541146e+09 | 2.38865613e+09 |
| R037 | practical-bf16-20261014-135107-82168ad5 / completed | 2.12226765e+09 | 2.1537751e+09 |
| R038 | practical-bf16-20261014-135107-82168ad5 / completed | 2.06249114e+09 | 2.15587226e+09 |
| R039 | practical-bf16-20261014-135107-82168ad5 / completed | 2.33470566e+09 | 2.38865613e+09 |

| Row ID | Run / state | Allocated bytes | Reserved bytes |
|---|---|---|---|
| R040 | practical-bf16-20261014-135107-82168ad5 / completed | 2.07541146e+09 | 2.38865613e+09 |
| R041 | practical-bf16-20261014-135107-82168ad5 / completed | 2.12226765e+09 | 2.1537751e+09 |
| R042 | practical-bf16-20261014-135107-82168ad5 / completed | 2.06249114e+09 | 2.15587226e+09 |
| R043 | practical-bf16-20261014-135107-82168ad5 / completed | 2.33470566e+09 | 2.38865613e+09 |
| R044 | practical-bf16-20261014-135107-82168ad5 / completed | 2.07541146e+09 | 2.38865613e+09 |
| R045 | practical-bf16-20261014-135107-82168ad5 / completed | 2.12226765e+09 | 2.1537751e+09 |
| R046 | practical-bf16-20261014-135107-82168ad5 / completed | 2.06249114e+09 | 2.15587226e+09 |
| R047 | practical-bf16-20261014-135107-82168ad5 / completed | 2.33470566e+09 | 2.38865613e+09 |
| R048 | practical-bf16-20261014-135107-82168ad5 / completed | 2.07541146e+09 | 2.38865613e+09 |

## Practical measurement scope

The historical FP16/INT8 stage's hardware-gates.json, when present, uses the same seed's FP16 whitening reference for both FP16 and INT8 solver factors. Its threshold concerns median latency and peak allocated memory only; it is not a same-bit INT8 gate and does not test p95, reserved memory, serialized bytes or quality. The later user-authorized BF16-only cohort is separate: seeds 11/23, five expected conditions, no other model formats and no three-seed promotion claim. BF16 has 16 storage bits but must not be labeled FP16. Its calibration routine saves full-target aggregate NLL and ECE, not per-block losses; practical NLL points are descriptive, without paired block intervals. Historical main-quality cohorts retain their separate block uncertainty. The separate descriptive ratios below use same-run, same-seed, same-bit whitening/anchor records with matching device, declared placement, compute precision and workload. Missing or ambiguous pairs remain excluded in report.json. No original hardware gate is relabeled.

Timings use batch size 1, 10 warmups and 30 synchronized samples. Cached 128-token decode timing excludes the initial prompt pass; prefill is measured separately at 128 and 512 tokens. Peak allocated/reserved bytes are CUDA allocator statistics including resident model and cache, not total driver GPU memory or system RAM. INT8 factors are dequantized for floating-point multiplication, so these timings do not establish native INT8-kernel acceleration.

Whole-model reload parity covers a 128-token prompt and one cached next-token forward, not the complete evaluation corpus or arbitrary generation. The historical dense-teacher file lacks this probe; the BF16-only amendment probes its dense teacher as well as its compressed models. Historical local factor probes and BF16 whole-model probes are distinct. The practical stage cap is cooperative; terminal elapsed wall time remains the charged cost.

## Same-bit descriptive cost ratios

Each ratio is anchor divided by matched whitening; below 1 is lower cost. Placement is explicit row metadata where available, otherwise the full-model resident device declared in the same-run configuration. These are exploratory measured workload comparisons, not new promotion gates.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 4; match by Row ID.

| Row ID | Run | Seed | Factor bits | Device / placement |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 | 11 | 16 | cuda / "resident" |
| R002 | practical-20261014-113642-0fe078b5 | 11 | 16 | cuda / "resident" |
| R003 | practical-20261014-113642-0fe078b5 | 11 | 16 | cuda / "resident" |
| R004 | practical-20261014-113642-0fe078b5 | 11 | 16 | cuda / "resident" |

| Row ID | Run | Seed | Factor bits | Device / placement |
|---|---|---|---|---|
| R005 | practical-20261014-113642-0fe078b5 | 11 | 8 | cuda / "resident" |
| R006 | practical-20261014-113642-0fe078b5 | 11 | 8 | cuda / "resident" |
| R007 | practical-20261014-113642-0fe078b5 | 11 | 8 | cuda / "resident" |
| R008 | practical-20261014-113642-0fe078b5 | 11 | 8 | cuda / "resident" |
| R009 | practical-bf16-20261014-135107-82168ad5 | 11 | 16 | cuda / "resident" |
| R010 | practical-bf16-20261014-135107-82168ad5 | 11 | 16 | cuda / "resident" |
| R011 | practical-bf16-20261014-135107-82168ad5 | 11 | 16 | cuda / "resident" |
| R012 | practical-bf16-20261014-135107-82168ad5 | 11 | 16 | cuda / "resident" |
| R013 | practical-bf16-20261014-135107-82168ad5 | 23 | 16 | cuda / "resident" |
| R014 | practical-bf16-20261014-135107-82168ad5 | 23 | 16 | cuda / "resident" |
| R015 | practical-bf16-20261014-135107-82168ad5 | 23 | 16 | cuda / "resident" |
| R016 | practical-bf16-20261014-135107-82168ad5 | 23 | 16 | cuda / "resident" |

Panel 2 of 4; match by Row ID.

| Row ID | Run | Mode | Prefill | Decode |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 | prefill | 128 | 0 |
| R002 | practical-20261014-113642-0fe078b5 | decode | 128 | 128 |
| R003 | practical-20261014-113642-0fe078b5 | prefill | 512 | 0 |
| R004 | practical-20261014-113642-0fe078b5 | decode | 512 | 128 |
| R005 | practical-20261014-113642-0fe078b5 | prefill | 128 | 0 |
| R006 | practical-20261014-113642-0fe078b5 | decode | 128 | 128 |
| R007 | practical-20261014-113642-0fe078b5 | prefill | 512 | 0 |
| R008 | practical-20261014-113642-0fe078b5 | decode | 512 | 128 |
| R009 | practical-bf16-20261014-135107-82168ad5 | prefill | 128 | 0 |
| R010 | practical-bf16-20261014-135107-82168ad5 | decode | 128 | 128 |
| R011 | practical-bf16-20261014-135107-82168ad5 | prefill | 512 | 0 |
| R012 | practical-bf16-20261014-135107-82168ad5 | decode | 512 | 128 |
| R013 | practical-bf16-20261014-135107-82168ad5 | prefill | 128 | 0 |
| R014 | practical-bf16-20261014-135107-82168ad5 | decode | 128 | 128 |
| R015 | practical-bf16-20261014-135107-82168ad5 | prefill | 512 | 0 |

| Row ID | Run | Mode | Prefill | Decode |
|---|---|---|---|---|
| R016 | practical-bf16-20261014-135107-82168ad5 | decode | 512 | 128 |

Panel 3 of 4; match by Row ID.

| Row ID | Run | Median ratio | p95 ratio | Allocated ratio |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 | 1.02320019 | 0.846438198 | 1 |
| R002 | practical-20261014-113642-0fe078b5 | 1.02734769 | 1.03668942 | 1 |
| R003 | practical-20261014-113642-0fe078b5 | 1.08332146 | 1.12398634 | 1 |
| R004 | practical-20261014-113642-0fe078b5 | 1.011668 | 0.983411217 | 1 |
| R005 | practical-20261014-113642-0fe078b5 | 1.03065529 | 0.798869532 | 1 |
| R006 | practical-20261014-113642-0fe078b5 | 1.01101759 | 1.01435857 | 1 |
| R007 | practical-20261014-113642-0fe078b5 | 1.03094508 | 1.03131513 | 1 |
| R008 | practical-20261014-113642-0fe078b5 | 1.0032714 | 1.02397524 | 1 |
| R009 | practical-bf16-20261014-135107-82168ad5 | 1.18761092 | 1.10590958 | 1 |
| R010 | practical-bf16-20261014-135107-82168ad5 | 1.10726922 | 1.16663946 | 1 |
| R011 | practical-bf16-20261014-135107-82168ad5 | 1.01079284 | 1.00785486 | 1 |
| R012 | practical-bf16-20261014-135107-82168ad5 | 0.987636065 | 0.974671384 | 1 |
| R013 | practical-bf16-20261014-135107-82168ad5 | 0.936992478 | 0.958638693 | 1 |
| R014 | practical-bf16-20261014-135107-82168ad5 | 0.967885422 | 0.936167024 | 1 |
| R015 | practical-bf16-20261014-135107-82168ad5 | 0.947309605 | 0.941740756 | 1 |
| R016 | practical-bf16-20261014-135107-82168ad5 | 0.957953587 | 0.928407146 | 1 |

Panel 4 of 4; match by Row ID.

| Row ID | Run | Reserved ratio | Factor-byte ratio | Whole-byte ratio |
|---|---|---|---|---|
| R001 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |
| R002 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |
| R003 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |
| R004 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |
| R005 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |
| R006 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |
| R007 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |
| R008 | practical-20261014-113642-0fe078b5 | 1 | 1 | 1 |

| Row ID | Run | Reserved ratio | Factor-byte ratio | Whole-byte ratio |
|---|---|---|---|---|
| R009 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |
| R010 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |
| R011 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |
| R012 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |
| R013 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |
| R014 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |
| R015 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |
| R016 | practical-bf16-20261014-135107-82168ad5 | 1 | 1 | 1 |

# Statistical interpretation

NLL differences use 2,000 paired bootstrap resamples of fixed contiguous 4,096-target blocks. The final partial block remains in full-test NLL but is excluded from fixed-block resampling. Individual dependent tokens are not bootstrapped. Per-seed confidence intervals, exact excluded counts, matched target checks and all scoped candidate-minus-baseline differences are in report.json and report.md. These are evaluation-block intervals, not seed intervals.

# Limitations

WikiText-2 has already informed this project, so these results are regression evidence rather than fresh-domain confirmation. No downstream-task capability or arbitrary-generation equivalence was measured. Only attention-output projections are compressed; hybrid DeltaNet/convolution mixers and all MLPs remain dense.

Within-model comparisons match factor budgets and native evaluation targets. Differences across model cohorts cannot be attributed solely to architecture: pretrained weights, tokenizers and, for the historical Gemma reference, inference precision differ. Raw perplexities must not rank different tokenizers. The later BF16 Gemma practical cohort measures costs and calibration; it does not add new BF16 drift evidence.

The original strict numerical-readiness failure remains binding. The two-seed BF16 practical amendment cannot satisfy a three-seed replication gate. Residual drift is a diagnostic proxy, not a mathematical certificate of global computation; lower perplexity on one corpus likewise does not establish preservation. The joint criterion is an operational research gate, not a universal theorem.

Weight-count reduction, actual serialized bytes and measured runtime are separate outcomes. Timings are descriptive for the recorded GPU/software/workloads, and reload probes cover only the stated prompt and cached token. Budget-limited frontier points remain unmeasured. Reboot accounting, including an unbounded first-reboot log tail, prevents an exact 24 GPU-hour compliance claim.

# Reproducibility and incomplete-run appendix

The accompanying report.json embeds each retained contract, source hashes, package/device versions, asset revisions, calibration/split hashes, target recipes, run-specific command, raw per-seed metrics, integrity checks, artifact paths and all failed/interrupted attempts. Read errors are retained explicitly; promotion is disabled.

| Stage | Recorded charged wall seconds | Declared cap seconds | Unknown-cost attempts |
|---|---|---|---|
| readiness | 1381.392 | 3600 | [] |
| regularization | 1589.148 | 14400 | [] |
| baselines | 3736.553 | 25200 | [] |
| composition | 12825.574 | 14400 | [] |
| architecture | 10720.138 | 14400 | [] |
| practical | 11524.515 | 14400 | [] |

## Preserved interrupted attempts and reused observations

The main architecture table and figures show the unique validated continuation, including its actual incomplete/terminal state. The source observations are identical reused measurements, not new replications. Original attempts and full metrics remain separately recorded in report.json and report.md; all source hashes and validation warnings are in presentation_supersession.

| Original attempt / state | Continuation / state | Reused rows | Original coverage | Charged wall seconds |
|---|---|---|---|---|
| architecture-replication-20261014-082807-5c73f669 / interrupted | architecture-replication-20261014-093845-97378b5e / incomplete | 10 | {"frontier_validation_records": 2, "note": "Record-count checks do not replace seed, target, recipe and provenance checks; an earlier 8-record cohort has no amended frontier evidence.", "primary_test_records": 8, "record_counts_complete": false, "required_frontier_validation_records": 18, "required_primary_test_records": 8, "validation_rank_divisors": [2]} | 3647.15864 |

Source provenance (architecture-replication-20261014-082807-5c73f669): {"config.json": "d9a0760000b6e7222b9671a5e0561d61bb9f361355fe777766d5189dcead9898", "contract.json": "703af918c7287806089ad2ec358b9df623af24552ce7b580e0c6fd462a680ab9", "metrics.json": "29422a72a2536d51795bd97abfb112e2a53d2957bed9e0b8be0bcab49aa4c621", "readiness.json": "067bae33ea5bd31a127520d46deadc34a79504cd987a948fd052e7aca31c50f3", "status.json": "2f66cc74973ccbf05b72874c6500e8b742ee6ad8d80770d52b2b81f10d8c32fd", "validation-teacher.json": "057c49abf6c7728756cf3acd2af6f03163b416d9195be9df34fff95cb88aeeb8"}.

Elapsed-wall-time charges include failed/retried work and may include reboot downtime. The first reboot has an unbounded unlogged tail; no exact GPU-hour compliance claim.

# Conclusion

Low-rank factors reduce targeted weight counts. Anchoring has no established consistent advantage over matched whitening on both likelihood and representation drift; reliable preservation of global computation remains unresolved. Recorded storage and runtime tradeoffs are reported separately, and all findings remain exploratory.

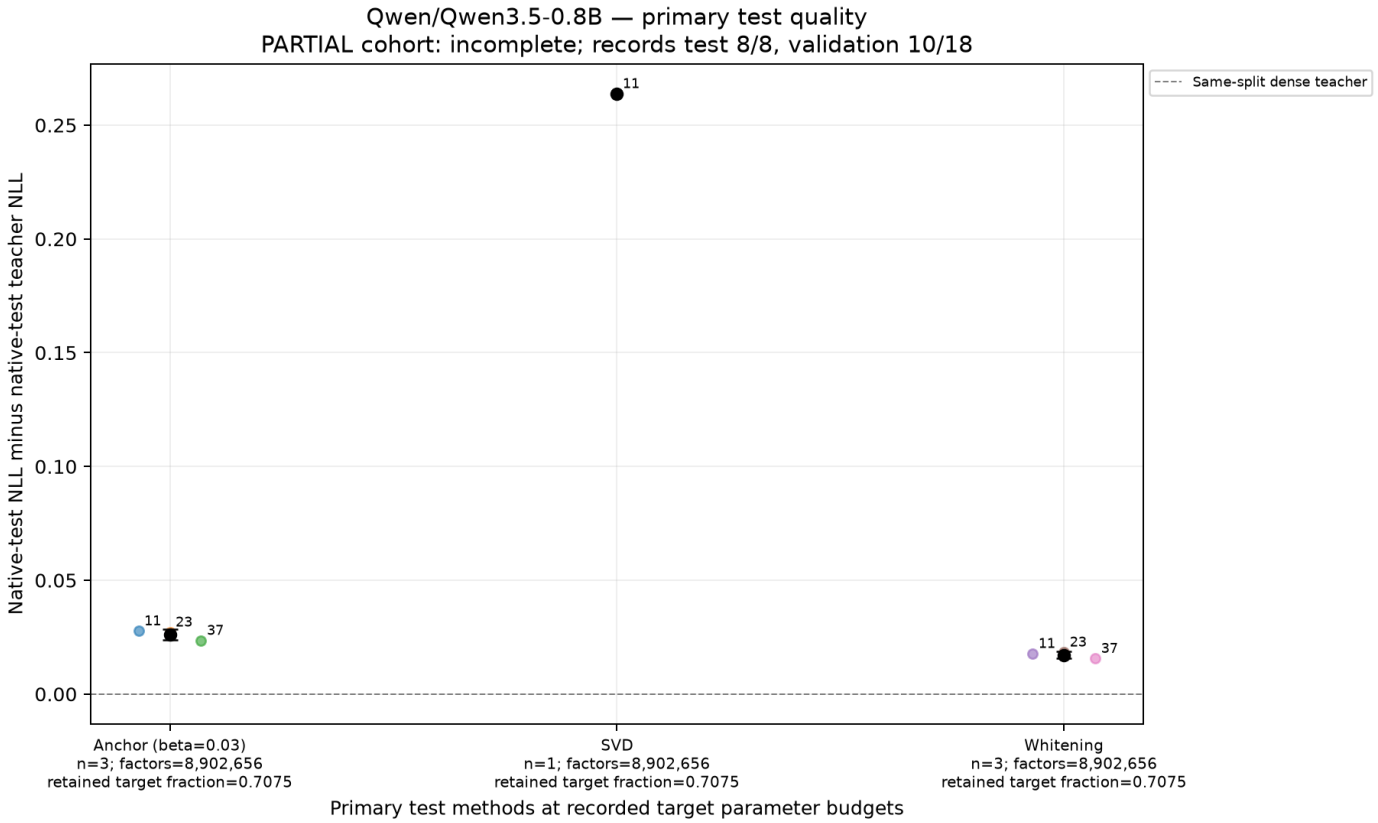
Complete negative or mixed comparisons are retained as bounded evidence. Unmeasured downstream tasks, fresh-corpus confirmation, absent seeds or hardware conditions are not implied by dispatcher completion.

# Verified bibliography and dates

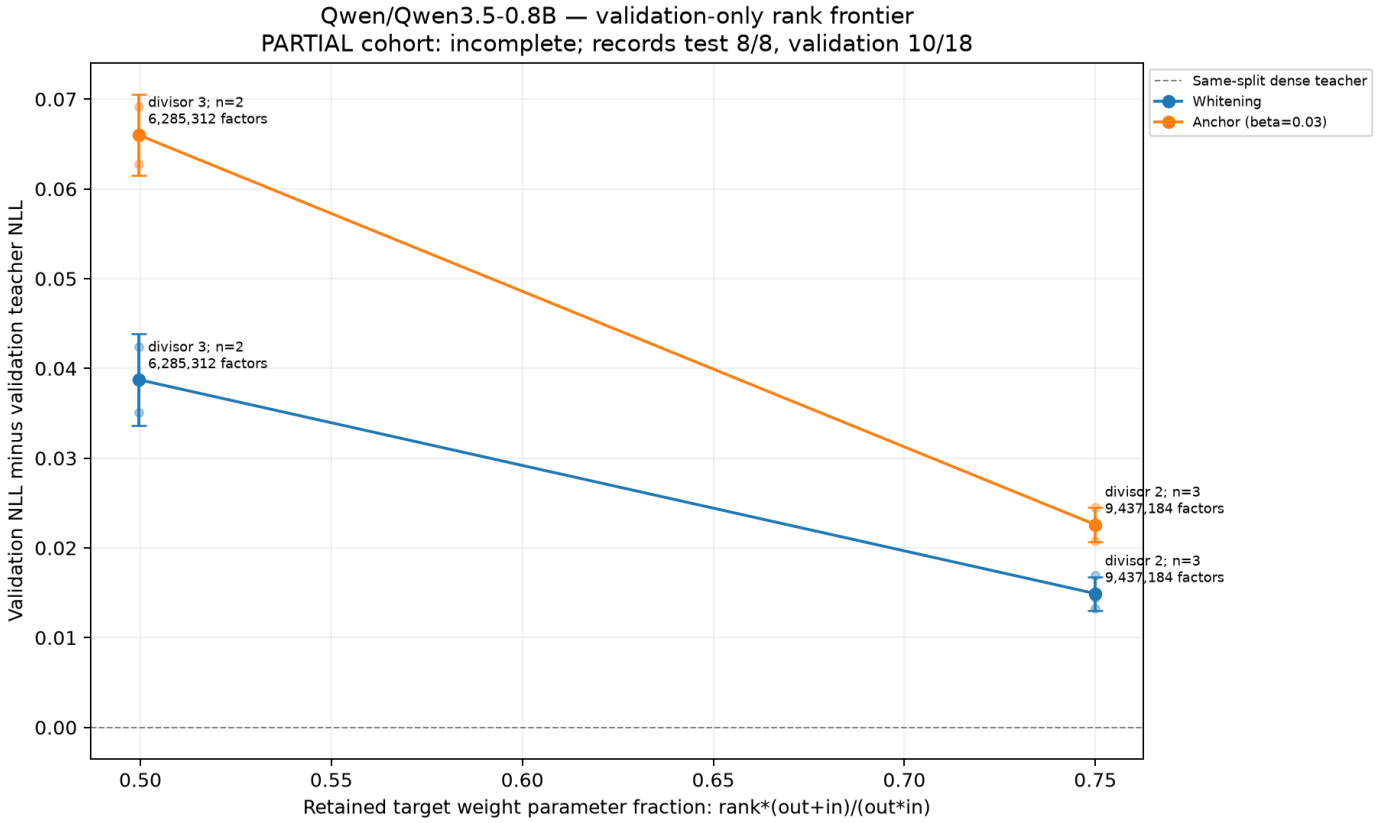
The additive BibTeX file (paper_protocol_v2.bib) contains the 22 plan references, verified from primary metadata on 2026-10-13. Preprint entries use the first submission year, record the checked revision, and separately note a venue when the primary page identifies one. In particular SVD-LLM is a 2024 preprint with ICLR 2025 metadata, GPTQ is a 2022 preprint with ICLR 2023 metadata, AWQ is a 2023 preprint with MLSys 2024 metadata, and SVDQuant is a 2024 preprint with ICLR 2025 metadata. EoRA's current primary metadata identifies ICLR 2026 workshops, not main-conference publication. T5 is cited as its 2020 JMLR article, with its 2019 preprint date recorded separately. Unverified venues are omitted.

Author lists reflect the verified source version. The ASVD v5 author list and the EoRA v6 list differ from some earlier bibliographies; this file does not silently mix versions. Hsu et al.'s author spelling follows the conference PDF (Sung-En Chang); the arXiv metadata spells that name Sungen Chang. The direct OpenReview page returned a browser challenge, so metadata was cross-checked with the primary paper's indexed PDF and its arXiv record.

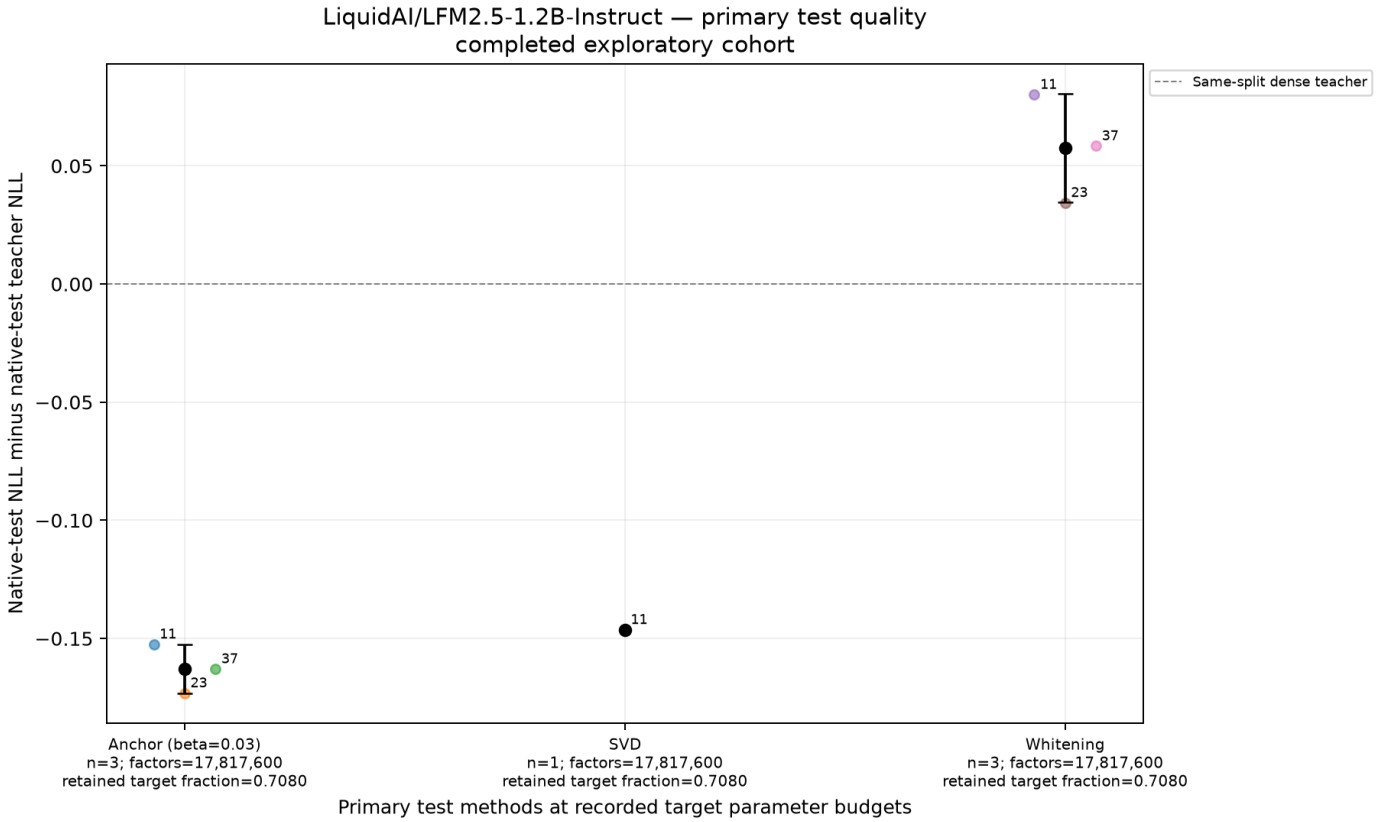
# Measured scientific figures



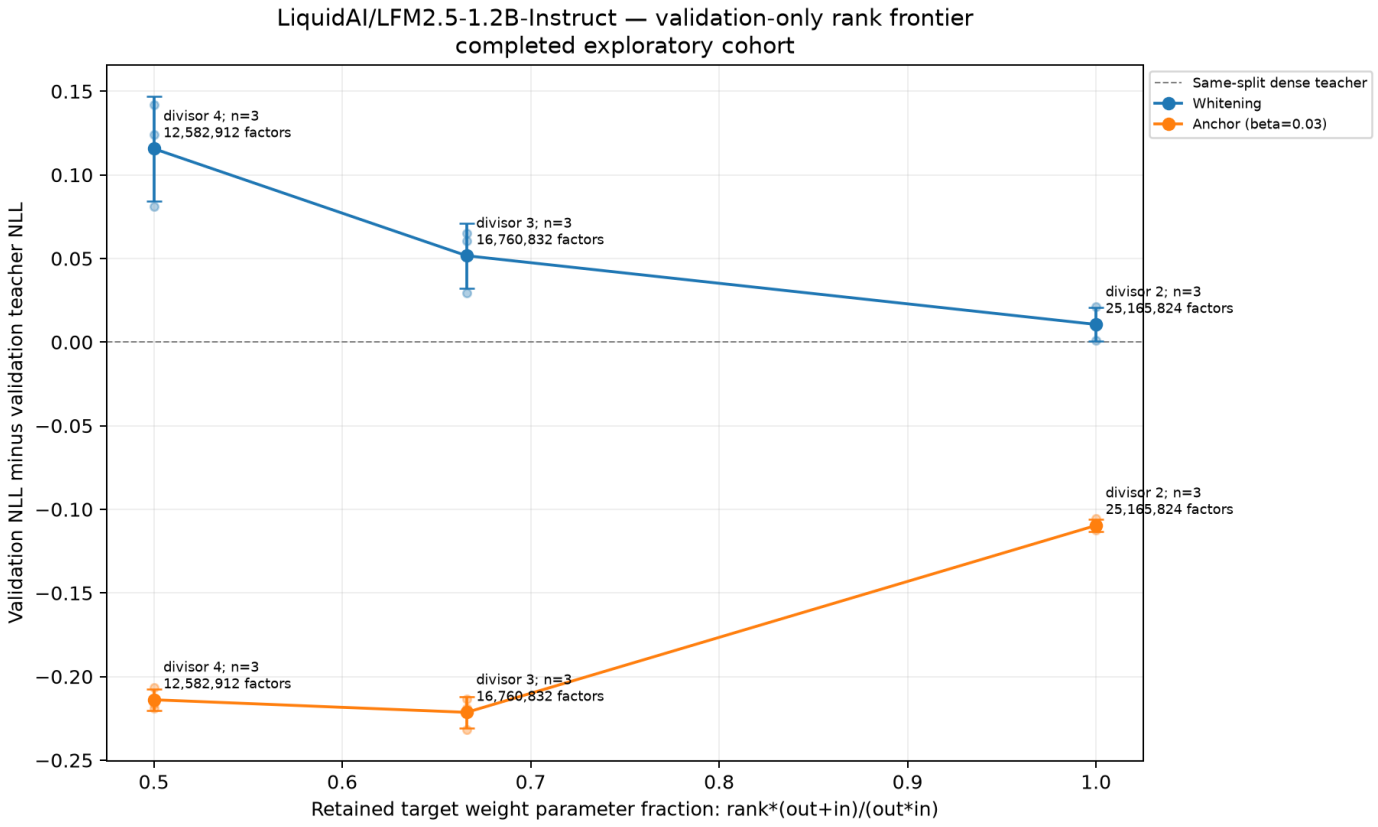

Qwen/Qwen3.5-0.8B: primary native-test teacher-relative NLL for PARTIAL cohort: incomplete; records test 8/8, validation 10/18; individual paired seed points, mean and sample seed SD when at least two seeds exist. Actual target factor counts/fractions shown. Strict promotion remains disabled.



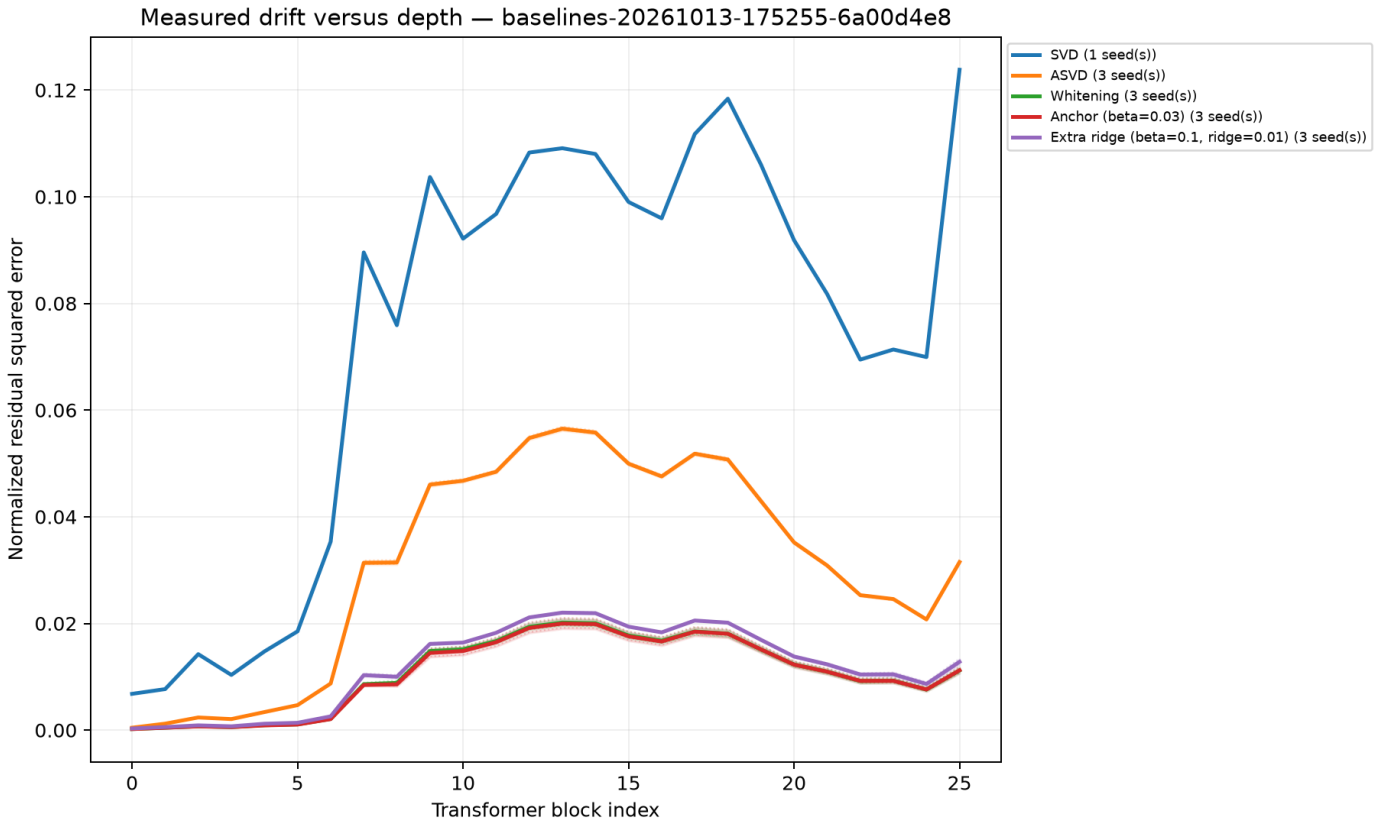

Qwen/Qwen3.5-0.8B: validation-only rank frontier for PARTIAL cohort: incomplete; records test 8/8, validation 10/18. Each point aggregates only one exact rank recipe and available paired seeds; error bars are sample seed SD. X is actual retained target weight parameter fraction, not whole-model storage or a rank compression factor. Validation is not independent confirmation.



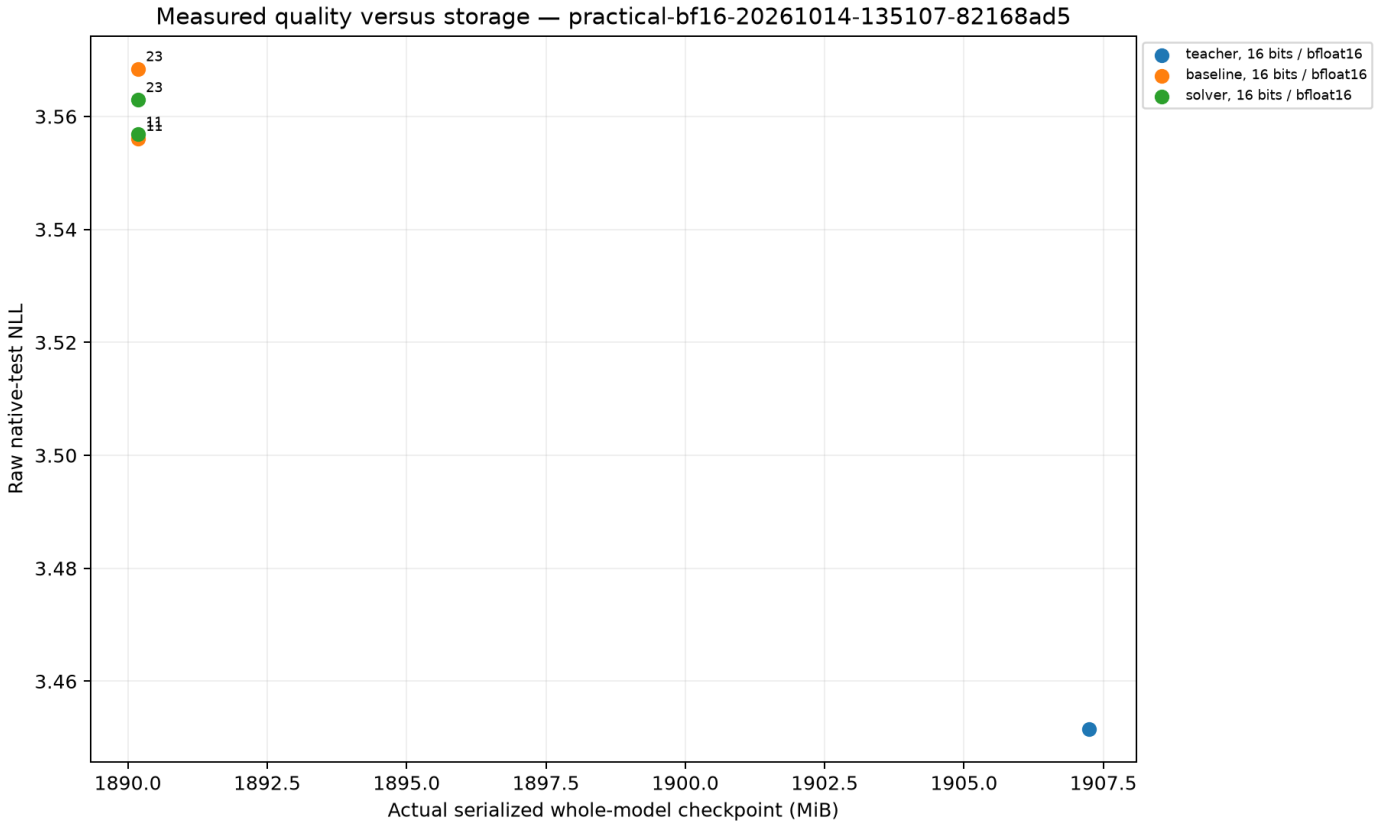

LiquidAI/LFM2.5-1.2B-Instruct: primary native-test teacher-relative NLL for completed exploratory cohort; individual paired seed points, mean and sample seed SD when at least two seeds exist. Actual target factor counts/fractions shown. Strict promotion remains disabled.




LiquidAI/LFM2.5-1.2B-Instruct: validation-only rank frontier for completed exploratory cohort. Each point aggregates only one exact rank recipe and available paired seeds; error bars are sample seed SD. X is actual retained target weight parameter fraction, not whole-model storage or a rank compression factor. Validation is not independent confirmation.




Gemma Stage C residual drift: individual seeds (dotted), mean (solid), sample seed SD (shaded). Single retained cohort; strict FP16 readiness remains failed.




Gemma Stage F raw native-test NLL versus actual whole-model checkpoint bytes; seed numbers label points. Factor-only full-rank controls excluded; no speed or lossless-compression claim.

Figure inputs, scope, source hashes and unavailable measurements are recorded in figures/provenance.json.

# Review export formats

Standalone LaTeX source and a separately generated ReportLab PDF are review artifacts. Their preparation status and source hashes are recorded in export-status.json. The ReportLab PDF is not compiled from LaTeX; successful TeX compilation requires a separate compiler diagnostic.

# Paired-block uncertainty appendix

Intervals are candidate minus the same-split baseline, using evaluation blocks rather than individual tokens. Partial-tail targets remain included in full NLL and are excluded from fixed-block intervals. These are not calibration-seed intervals.

## architecture-replication-20261014-082807-5c73f669

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {"rank": 483, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.24604289213573946 | 0.24615020975613225 | [0.24006466116265526, 0.25237916880828776] | 72/2000 | 2280 |

## architecture-replication-20261014-082807-5c73f669

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 483, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.009941660681098283 | 0.010034525719555844 | [0.00827384469021905, 0.01189700522595723] | 72/2000 | 2280 |
| 23 | 0.008791898301914979 | 0.008937712051082021 | [0.0072091521283551875, 0.010677830953140406] | 72/2000 | 2280 |
| 37 | 0.007921984778739719 | 0.007972056491898503 | [0.006199138578762092, 0.009875832550118127] | 72/2000 | 2280 |

## architecture-replication-20261014-082807-5c73f669

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 512, "rank_divisor": 2, "split": "validation", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.011256371692945919 | 0.011285927374891085 | [0.00961544286242256, 0.013039805029695348] | 63/2000 | 3362 |

## architecture-replication-20261014-093845-97378b5e

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {"rank": 483, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.23.self_attn.o_proj",

"precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.24604289213573946 | 0.24615020975613225 | [0.24006466116265526, 0.25237916880828776] | 72/2000 | 2280 |

## architecture-replication-20261014-093845-97378b5e

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 483, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.009941660681098283 | 0.010034525719555844 | [0.00827384469021905, 0.01189700522595723] | 72/2000 | 2280 |
| 23 | 0.008791898301914979 | 0.008937712051082021 | [0.0072091521283551875, 0.010677830953140406] | 72/2000 | 2280 |
| 37 | 0.007921984778739719 | 0.007972056491898503 | [0.006199138578762092, 0.009875832550118127] | 72/2000 | 2280 |

## architecture-replication-20261014-093845-97378b5e

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 512, "rank_divisor": 2, "split": "validation", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1572864, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 512, "shape": [1024, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.011256371692945919 | 0.011285927374891085 | [0.00961544286242256, 0.013039805029695348] | 63/2000 | 3362 |
| 23 | 0.005501045557622271 | 0.005585379198455225 | [0.0038466850466807364, 0.007429576815721795] | 63/2000 | 3362 |
| 37 | 0.006284480060781636 | 0.006343413931392497 | [0.004643602466799244, 0.008036439451197867] | 63/2000 | 3362 |

## architecture-replication-20261014-093845-97378b5e

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 341, "rank_divisor": 3, "split": "validation", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1047552, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 341, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1047552, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 341, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1047552, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 341, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1047552, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 341, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1047552, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 341, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1047552, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 341, "shape": [1024, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.027697128973382466 | 0.027779999025171056 | [0.025335660561068965, 0.030283496110847567] | 63/2000 | 3362 |
| 23 | 0.026820453601597936 | 0.027022706044184327 | [0.024655860298807966, 0.02943987465715239] | 63/2000 | 3362 |

## architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {"rank": 725, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304,

"factor_parameters": 2969600, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.22646344793647272 | -0.2264399873327979 | [-0.24367673627667688, -0.20951701448693727] | 73/2000 | 24 |

## architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 725, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.23260833182808122 | -0.23258001440330633 | [-0.24672731477834628, -0.21835409049405202] | 73/2000 | 24 |
| 23 | -0.20752383848706338 | -0.20750846944987744 | [-0.2240134011794247, -0.19023832839560784] | 73/2000 | 24 |
| 37 | -0.22133336282727098 | -0.2213271375686348 | [-0.23908078459653 28, -0.2034092545763355] | 73/2000 | 24 |

## architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 1024, "rank_divisor": 2, "split": "validation", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.1267830725542116 | -0.1264940554545641 | [-0.13380107717694656, -0.11900781495134909] | 63/2000 | 3770 |
| 23 | -0.11213960954857505 | -0.1108899550184011 | [-0.11815167612563775, -0.10308410012643247] | 63/2000 | 3770 |
| 37 | -0.12168323184273745 | -0.12028804990045704 | [-0.12827400810641157, -0.11168668095270724] | 63/2000 | 3770 |

## architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 682, "rank_divisor": 3, "split": "validation", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.27847398152028663 | -0.2766533448376106 | [-0.2934387856600625, -0.25966835384794923] | 63/2000 | 3770 |
| 23 | -0.2610503911613548 | -0.25815498372455165 | [-0.2751110691947673, -0.24063632729316714] | 63/2000 | 3770 |
| 37 | -0.2795701052254451 | -0.27724325879898043 | [-0.29641486111395143, -0.2577876479611187] | 63/2000 | 3770 |

## architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 512, "rank_divisor": 4, "split": "validation", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}]}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.34814383348227884 | -0.3461968847176596 | [-0.36750653404331235, -0.3257971909558875] | 63/2000 | 3770 |
| 23 | -0.29971982483469123 | -0.2967208833552329 | [-0.31580071090051415, -0.2776262464332278] | 63/2000 | 3770 |
| 37 | -0.3406797660259072 | -0.3379485954229951 | [-0.36191847249031006, -0.313945935288103] | 63/2000 | 3770 |

## baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 1.5259270359038637 | 1.526963313415228 | [1.4943535833878134, 1.5596674649359294] | 71/2000 | 1465 |

## baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "asvd"}, "scope": {}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.13435744983933437 | 0.13448046045923576 | [0.12208941117886445, 0.14684303527631282] | 71/2000 | 1465 |
| 23 | 0.12406477628810314 | 0.12440374314135588 | [0.1140738804768457, 0.13439828040507174] | 71/2000 | 1465 |
| 37 | 0.10653445125745864 | 0.10667353359540374 | [0.09858797073867313, 0.11492986771691276] | 71/2000 | 1465 |

## baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0031192516557841613 | -0.003069222128221489 | [-0.005598433325437209, -0.0005771062229211223] | 71/2000 | 1465 |
| 23 | -0.00867016563638367 | -0.008560964606249673 | [-0.01236161516313488, -0.004879390368432393] | 71/2000 | 1465 |
| 37 | -0.008888696495308254 | -0.008889474302069626 | [-0.011564302129920533, -0.006193025270745394] | 71/2000 | 1465 |

## baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01}, "scope": {}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.007341743295384272 | 0.007358043732914079 | [0.003578168563012017, 0.011322591469446464] | 71/2000 | 1465 |
| 23 | 0.010085729923213327 | 0.010195171973247596 | [0.005499654363179419, 0.01504279117529553] | 71/2000 | 1465 |
| 37 | 0.0012338436455246004 | 0.0012706044011388724 | [-0.002438464246056019, 0.00520269695332208] | 71/2000 | 1465 |

## composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -4.60899341388781e-05 | -4.0872599009493115e-05 | [-0.000217734233338580 93, 0.0001685818120544 4938] | 71/2000 | 1465 |
| 23 | -3.271375526325137e-05 | -3.199588131774163e-05 | [-0.000174187935932211 99, 0.0001136384687997568] | 71/2000 | 1465 |
| 37 | -6.647979225569334e-05 | -7.095429546745745e-05 | [-0.000291885207220884 4, 9.73636943746164e-05] | 71/2000 | 1465 |

## composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0008545763004039009 | -0.0008561104717573092 | [-0.001150257172183655 8, -0.00056394938574460 14] | 71/2000 | 1465 |
| 23 | -9.84808364817539e-05 | -9.306537422449584e-05 | [-0.000375805062010882 6, 0.00017279865033056 432] | 71/2000 | 1465 |
| 37 | -7.626569138096745e-05 | -7.009637664775361e-05 | [-0.000360900941821140 16, 0.00023415362379323] | 71/2000 | 1465 |

## composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.0001729304994845724 8 | 0.0001719993236095868 5 | [4.463885584017398e-05, 0.0002936832496752188] | 71/2000 | 1465 |
| 23 | -0.0007079791885041549 | -0.000699445138327318 | [-0.000889761639764939 4, -0.000521697610992854] | 71/2000 | 1465 |
| 37 | -7.243953146688717e-05 | -6.619463365792254e-05 | [-0.000212401684606083 8, 7.996392602977384e-05] | 71/2000 | 1465 |

## composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0002764577707105269 | -0.000305017713926657 | [-0.000801674596175575 1, 0.00016777266329458 292] | 71/2000 | 1465 |
| 23 | 0.0002748313234466563 5 | 0.0002560327026930727 | [-0.000328662642760605 2, 0.0008294795855742868] | 71/2000 | 1465 |
| 37 | -0.0001416761346267915 2 | -0.0001526383887316477 5 | [-0.000643328925356078 7, 0.0003204156261228786] | 71/2000 | 1465 |

## composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0005983137392528166 | -0.0005957751251080846 | [-0.000967770908744049 3, -0.00022507084570211 577] | 71/2000 | 1465 |
| 23 | 0.0004492915974461228 | 0.0004649281307566705 5 | [5.583698743252717e-05, 0.0008733775102226093] | 71/2000 | 1465 |
| 37 | 0.0001664124767705408 6 | 0.0001822159329953418 2 | [-0.000260722413231894 8, 0.000609754279693661] | 71/2000 | 1465 |

## composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 8.042555401250508e-05 | 8.104170434346619e-05 | [-7.955390526647082e-05 , 0.000235949191167862 98] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -4.60899341388781e-05 | -4.0872599009493115e-0 5 | [-0.000217734233338580 93, 0.0001685818120544 4938] | 71/2000 | 1465 |
| 23 | -3.271375526325137e-05 | -3.199588131774163e-05 | [-0.000174187935932211 99, 0.0001136384687997568] | 71/2000 | 1465 |
| 37 | -6.647979225569334e-05 | -7.095429546745745e-05 | [-0.000291885207220884 4, 9.73636943746164e-05] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0008545763004039009 | -0.0008561104717573092 | [-0.001150257172183655 8, -0.00056394938574460 14] | 71/2000 | 1465 |
| 23 | -9.84808364817539e-05 | -9.306537422449584e-05 | [-0.000375805062010882 6, 0.00017279865033056 432] | 71/2000 | 1465 |
| 37 | -7.626569138096745e-05 | -7.009637664775361e-05 | [-0.000360900941821140 16, 0.00023415362379323] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.0001729304994845724 8 | 0.0001719993236095868 5 | [4.463885584017398e-05, 0.0002936832496752188] | 71/2000 | 1465 |
| 23 | -0.0007079791885041549 | -0.000699445138327318 | [-0.000889761639764939 4, -0.000521697610992854] | 71/2000 | 1465 |
| 37 | -7.243953146688717e-05 | -6.619463365792254e-05 | [-0.000212401684606083 8, 7.996392602977384e-05] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0002764577707105269 | -0.000305017713926657 | [-0.000801674596175575 1, 0.00016777266329458292] | 71/2000 | 1465 |
| 23 | 0.0002748313234466563 5 | 0.0002560327026930727 | [-0.000328662642760605 2, 0.0008294795855742868] | 71/2000 | 1465 |
| 37 | -0.0001416761346267915 2 | -0.0001526383887316477 5 | [-0.000643328925356078 7, 0.0003204156261228786] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0005983137392528166 | -0.0005957751251080846 | [-0.000967770908744049 3, -0.00022507084570211577] | 71/2000 | 1465 |
| 23 | 0.0004492915974461228 | 0.0004649281307566705 5 | [5.583698743252717e-05, 0.0008733775102226093] | 71/2000 | 1465 |
| 37 | 0.0001664124767705408 6 | 0.0001822159329953418 2 | [-0.000260722413231894 8, 0.000609754279693661] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 8.042555401250508e-05 | 8.104170434346619e-05 | [-7.955390526647082e-05 , 0.000235949191167862 98] | 71/2000 | 1465 |
| 23 | -0.0008111692747406037 | -0.0008061970833859053 | [-0.001029544344414319 8, -0.00058981680664782 52] | 71/2000 | 1465 |
| 37 | 0.0001807916412591724 4 | 0.0001831437652353506 8 | [-9.780179775247661e-06 , 0.0003823935562813924] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0006571169305328121 | -0.0006716296099761055 | [-0.00164636577805492, 0.0001938152765730989 4] | 71/2000 | 1465 |
| 23 | 7.183583249946473e-05 | 6.899960868660811e-05 | [-0.000682006490905095 6, 0.0007963317650939533] | 71/2000 | 1465 |
| 37 | -0.0002972410869361785 3 | -0.0003128500272375454 3 | [-0.000933935280597861 5, 0.0002562151766575395] | 71/2000 | 1465 |

## composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [11, 12, 13, 14], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.003885308129297546 | 0.0039059187090884325 | [0.0026608101493276353, 0.0051295292904874204] | 71/2000 | 1465 |
| 23 | 0.003589996487403734 | 0.003661064528190402 | [0.002338269696115027, 0.004953754396441149] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -4.60899341388781e-05 | -4.0872599009493115e-05 | [-0.0002177342333385808 93, 0.0001685818120544 4938] | 71/2000 | 1465 |
| 23 | -3.271375526325137e-05 | -3.199588131774163e-05 | [-0.0001741879359322119 9, 0.0001136384687997568] | 71/2000 | 1465 |
| 37 | -6.647979225569334e-05 | -7.095429546745745e-05 | [-0.0002918852072208844, 9.73636943746164e-05] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0008545763004039009 | -0.0008561104717573092 | [-0.001150257172183655 8, -0.00056394938574460 14] | 71/2000 | 1465 |
| 23 | -9.84808364817539e-05 | -9.306537422449584e-05 | [-0.0003758050620108826, 0.00017279865033056 432] | 71/2000 | 1465 |
| 37 | -7.626569138096745e-05 | -7.009637664775361e-05 | [-0.0003609009418211401 16, 0.00023415362379323] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.0001729304994845724 8 | 0.0001719993236095868 5 | [4.463885584017398e-05, 0.0002936832496752188] | 71/2000 | 1465 |
| 23 | -0.0007079791885041549 | -0.000699445138327318 | [-0.000889761639764939 4, -0.000521697610992854] | 71/2000 | 1465 |
| 37 | -7.243953146688717e-05 | -6.619463365792254e-05 | [-0.0002124016846060838, 7.996392602977384e-05] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0002764577707105269 | -0.000305017713926657 | [-0.000801674596175575 1, 0.00016777266329458 292] | 71/2000 | 1465 |
| 23 | 0.0002748313234466563 5 | 0.0002560327026930727 | [-0.000328662642760605 2, 0.0008294795855742868] | 71/2000 | 1465 |

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 37 | -0.0001416761346267915 2 | -0.0001526383887316477 5 | [-0.000643328925356078 7, 0.0003204156261228786] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0005983137392528166 | -0.0005957751251080846 | [-0.000967770908744049 3, -0.00022507084570211 577] | 71/2000 | 1465 |
| 23 | 0.0004492915974461228 | 0.0004649281307566705 5 | [5.583698743252717e-05, 0.0008733775102226093] | 71/2000 | 1465 |
| 37 | 0.0001664124767705408 6 | 0.0001822159329953418 2 | [-0.000260722413231894 8, 0.000609754279693661] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 8.042555401250508e-05 | 8.104170434346619e-05 | [-7.955390526647082e-05 , 0.000235949191167862 98] | 71/2000 | 1465 |
| 23 | -0.0008111692747406037 | -0.0008061970833859053 | [-0.001029544344414319 8, -0.00058981680664782 52] | 71/2000 | 1465 |
| 37 | 0.0001807916412591724 4 | 0.0001831437652353506 8 | [-9.780179775247661e-06 , 0.0003823935562813924] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0006571169305328121 | -0.0006716296099761055 | [-0.00164636577805492, 0.0001938152765730989 4] | 71/2000 | 1465 |
| 23 | 7.183583249946473e-05 | 6.899960868660811e-05 | [-0.000682006490905095 6, 0.0007963317650939533] | 71/2000 | 1465 |
| 37 | -0.0002972410869361785 3 | -0.0003128500272375454 3 | [-0.000933935280597861 5, 0.0002562151766575395] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [11, 12, 13, 14], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | 0.003885308129297546 | 0.0039059187090884325 | [0.0026608101493276353 , 0.0051295292904874204] | 71/2000 | 1465 |
| 23 | 0.003589996487403734 | 0.003661064528190402 | [0.002338269696115027, 0.004953754396441149] | 71/2000 | 1465 |
| 37 | 0.0006958575765264769 | 0.000706815205475545 | [-0.000485747489089789 36, 0.0019332813322892252] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [22, 23, 24, 25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0004223128795064941 | -0.00041862580332443627 | [-0.000905342181576156, 8.412072718029896e-05] | 71/2000 | 1465 |
| 23 | -0.00290206343154077 | -0.0029018436309528315 | [-0.003464295129582087, -0.0023463511594901273] | 71/2000 | 1465 |
| 37 | 0.0003427221577525009 | 0.0003274063402903456 | [-0.000174796254365639, 0.000831103017138969] | 71/2000 | 1465 |

## composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"}}

| Seed | Full NLL delta | Subset delta | 95% interval | Blocks/resamples | Excluded targets |
|---|---|---|---|---|---|
| 11 | -0.0031192516557841613 | -0.003069222128221489 | [-0.005598433325437209, -0.0005771062229211223] | 71/2000 | 1465 |
| 23 | -0.00867016563638367 | -0.008560964606249673 | [-0.01236161516313488, -0.004879390368432393] | 71/2000 | 1465 |
| 37 | -0.008888696495308254 | -0.008889474302069626 | [-0.011564302129920533, -0.006193025270745394] | 71/2000 | 1465 |

# Verified references

[1] [chen2021drone] Patrick Chen and Hsiang-Fu Yu and Inderjit S. Dhillon and Cho-Jui Hsieh. DRONE: Data-aware Low-rank Compression for Large NLP Models. Advances in Neural Information Processing Systems. 2021. https://proceedings.neurips.cc/paper/2021/hash/f56de5ef149cf0aedcc8f4797031e229-Abstract.html

[2] [hsu2022weighted] Yen-Chang Hsu and Ting Hua and Sung-En Chang and Qian Lou and Yilin Shen and Hongxia Jin. Language Model Compression with Weighted Low-Rank Factorization. International Conference on Learning Representations. 2022. Author spelling from conference PDF; arXiv:2207.00112 first submitted 2022. https://openreview.net/pdf?id=uPv9Y3gmAI5

[3] [yuan2023asvd] Zhihang Yuan and Yuzhang Shang and Yue Song and Dawei Yang and Qiang Wu and Yan Yan and Guangyu Sun. ASVD: Activation-aware Singular Value Decomposition for Compressing Large Language Models. 2023. First submitted 10 December 2023; verified revision v5 dated 28 August 2025. https://arxiv.org/abs/2312.05821

[4] [wang2024svdllm] Xin Wang and Yu Zheng and Zhongwei Wan and Mi Zhang. SVD-LLM: Truncation-aware Singular Value Decomposition for Large Language Model Compression. 2024. First submitted 12 March 2024; arXiv metadata identifies ICLR 2025; verified revision v5. https://arxiv.org/abs/2403.07378

[5] [wang2025svdllmv2] Xin Wang and Samiul Alam and Zhongwei Wan and Hui Shen and Mi Zhang. SVD-LLM V2: Optimizing Singular Value Truncation for Large Language Model Compression. Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers). 2025. https://aclanthology.org/2025.naacl-long.217/

[6] [liu2024eora] Shih-Yang Liu and Maksim Khadkevich and Nai Chit Fung and Charbel Sakr and Chao-Han Huck Yang and Chien-Yi Wang and Saurav Muralidharan and Hongxu Yin and Kwang-Ting Cheng and Jan Kautz and Yu-Chiang Frank Wang and Pavlo Molchanov and Min-Hung Chen. EoRA: Fine-tuning-free Compensation for Compressed LLM with Eigenspace Low-Rank Approximation. 2024. First submitted 28 October 2024; verified v6 dated 17 March 2026; arXiv metadata says ICLR 2026 workshops, not main conference. https://arxiv.org/abs/2410.21271

[7] [hu2026saessvd] Xing Hu and Dawei Yang and Yuan Cheng and Zhixuan Chen and Zukang Xu. SAES-SVD: Self-Adaptive Suppression of Accumulated and Local Errors for SVD-based LLM Compression. 2026. First submitted 3 February 2026; preprint, venue not verified. https://arxiv.org/abs/2602.03051

[8] [liu2026lacesvd] Zhuowen Liu and Longkun Hao and Shiyu Feng and Xiaowen Chang and Ruiqun Li and Changqun Li. LACE-SVD: Loss-Aware SVD with Cumulative Error Correction for LLM Compression. 2026. First submitted 3 July 2026; verified v2 dated 15 August 2026; preprint, venue not verified. https://arxiv.org/abs/2607.03057

[9] [abbasi2026zerosum] Ali Abbasi and Chayne Thrash and Haoran Qin and Shansita Sharma and Sepehr Seifi and Soheil Kolouri. Zero Sum SVD: Balancing Loss Sensitivity for Low Rank LLM Compression. 2026. First submitted 2 February 2026; verified v2 dated 28 September 2026; preprint, venue not verified. https://arxiv.org/abs/2602.02848

[10] [abbasi2026iosvd] Ali Abbasi and Chayne Thrash and Haoran Qin and Hamed Pirsiavash and Soheil Kolouri. IO-SVD: Input-Output Whitened SVD for Adaptive-Rank LLM Compression. 2026. First submitted 15 May 2026; preprint, venue not verified. https://arxiv.org/abs/2605.15626

[11] [arai2025qep] Yamato Arai and Yuma Ichikawa. Quantization Error Propagation: Revisiting Layer-Wise Post-Training Quantization. 2025. First submitted 13 April 2025; arXiv metadata says accepted at NeurIPS 2025; verified v3 dated 13 January 2026. https://arxiv.org/abs/2504.09629

[12] [frantar2022gptq] Elias Frantar and Saleh Ashkboos and Torsten Hoefler and Dan Alistarh. GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. 2022. First submitted 31 October 2022; arXiv metadata identifies ICLR 2023; verified v2. https://arxiv.org/abs/2210.17323

[13] [lin2023awq] Ji Lin and Jiaming Tang and Haotian Tang and Shang Yang and Wei-Ming Chen and Wei-Chen Wang and Guangxuan Xiao and Xingyu Dang and Chuang Gan and Song Han. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. 2023. First submitted 1 June 2023; arXiv metadata identifies MLSys 2024; verified v6 dated 25 April 2026. https://arxiv.org/abs/2306.00978

[14] [frantar2023sparsegpt] Elias Frantar and Dan Alistarh. SparseGPT: Massive Language Models Can Be Accurately Pruned in One-Shot. 2023. First submitted 2 January 2023; verified v3; this entry cites the preprint. https://arxiv.org/abs/2301.00774

[15] [li2024svdquant] Muyang Li and Yujun Lin and Zhekai Zhang and Tianle Cai and Xiuyu Li and Junxian Guo and Enze Xie and Chenlin Meng and Jun-Yan Zhu and Song Han. SVDQuant: Absorbing Outliers by Low-Rank Components for 4-Bit Diffusion Models. 2024. First submitted 7 November 2024; arXiv metadata identifies ICLR 2025 Spotlight; verified v4; diffusion-model scope. https://arxiv.org/abs/2411.05007

[16] [kornblith2019cka] Simon Kornblith and Mohammad Norouzi and Honglak Lee and Geoffrey Hinton. Similarity of Neural Network Representations Revisited. Proceedings of the 36th International Conference on Machine Learning. 2019. https://proceedings.mlr.press/v97/kornblith19a.html

[17] [guo2017calibration] Chuan Guo and Geoff Pleiss and Yu Sun and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. Proceedings of the 34th International Conference on Machine Learning. 2017. https://proceedings.mlr.press/v70/guo17a.html

[18] [biderman2024evaluation] Stella Biderman and Hailey Schoelkopf and Lintang Sutawika and Leo Gao and Jonathan Tow and Baber Abbasi and Alham Fikri Aji and Pawan Sasanka Ammanamanchi and Sidney Black and Jordan Clive and Anthony DiPofi and Julen Etxaniz and Benjamin Fattori and Jessica Zosa Forde and Charles Foster and Jeffrey Hsu and Mimansa Jaiswal and Wilson Y. Lee and Haonan Li and Charles Lovering and Niklas Muennighoff and Ellie Pavlick and Jason Phang and Aviya Skowron and Samson Tan and Xiangru Tang and Kevin A. Wang and Genta Indra Winata and Fran\c cois Yvon and Andy Zou. Lessons from the Trenches on Reproducible Evaluation of Language Models. 2024. First submitted 23 May 2024; author list verified against v3 dated 31 May 2026; this entry cites the preprint. https://arxiv.org/abs/2405.14782

[19] [merity2016wikitext] Stephen Merity and Caiming Xiong and James Bradbury and Richard Socher. Pointer Sentinel Mixture Models. 2016. First submitted 26 September 2016; this entry cites the WikiText-introducing preprint. https://arxiv.org/abs/1609.07843

[20] [raffel2020t5] Colin Raffel and Noam Shazeer and Adam Roberts and Katherine Lee and Sharan Narang and Michael Matena and Yanqi Zhou and Wei Li and Peter J. Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research. 2020. Journal publication 2020; original arXiv:1910.10683 first submitted 23 October 2019. https://jmlr.org/papers/v21/20-074.html

[21] [zellers2019hellaswag] Rowan Zellers and Ari Holtzman and Yonatan Bisk and Ali Farhadi and Yejin Choi. HellaSwag: Can a Machine Really Finish Your Sentence?. Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. 2019. https://aclanthology.org/P19-1472/

[22] [clark2018arc] Peter Clark and Isaac Cowhey and Oren Etzioni and Tushar Khot and Ashish Sabharwal and Carissa Schoenick and Oyvind Tafjord. Think you have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge. 2018. First submitted 14 March 2018; this entry cites the preprint. https://arxiv.org/abs/1803.05457

# Source integrity

report.json SHA256 f49052f7ec6ade2797c90609f46714da7abdc7358c1e9962028fc6bb625c4b38

manuscript.md SHA256 bc876c820f0c7d62cf2b718cd8278fef922a47501b747b207b0e4902970de868

paper_protocol_v2.bib SHA256 e29efabec9975352119f60aafa5da509eaa2060ce7ba74ff7d0f7071afc49261

export_paper_pdf_20261014.py SHA256 4c683e7c6973222a22714d120928cb952165d216792a44e8a6956b93d0ca9deb